



CSE 326:INFORMATION SYSTEM DESIGN SESSIONAL

FINAL REPORT ON DIAGRAMS AND WEB APPLICATION

Udemy:Course Management System

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April 7, 2026

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Chapter 1

Introduction

1.1 Background

The rapid growth of online education has fundamentally changed the way knowledge is delivered and consumed across the globe. Platforms like Udemy have emerged as key players in this transformation, offering a vast ecosystem where instructors and learners interact through structured digital content. As the demand for scalable and feature-rich e-learning systems continues to rise, the need for well-designed course management systems becomes increasingly important.

1.2 About the Platform

Udemy is a global online learning platform hosting over 200,000 courses, trusted by millions of students worldwide. It provides instructors with tools to create and deliver high-quality educational content while offering learners flexible access to knowledge across diverse domains.

1.3 Project Overview

This report presents the design and implementation of a Course Management Module for Udemy, developed as part of the CSE 326: Information System Design Sessional. The project focuses on a selected subset of Udemy's core functionality, specifically the features that enable instructors to create, organize, and deliver engaging online courses with multimedia content. Basic authentication features such as login, registration, and

user profile management are intentionally kept outside the scope of this work.

1.4 Core Features

The module encompasses four core features:

- **Video-based Lessons and Playback** — structured video content delivery with playback controls and subtitle support.
- **Automatic Progress Tracking** — seamless monitoring of student progress with analytics and smart notifications.
- **Downloadable Learning Materials** — flexible resource management for offline access.
- **Course Rating and Review System** — authentic feedback mechanism fostering quality and engagement.

1.5 Assigned Feature

Among the four core features, our group has been specifically assigned the **Automatic Progress Tracking** feature. In any online learning environment, one of the biggest challenges is keeping students consistently engaged and motivated throughout a course. Progress tracking directly addresses this problem by automatically monitoring how much of a course a student has completed, sending timely reminders, and providing both students and instructors with meaningful insights through an analytics dashboard. Rather than leaving learners to manually keep track of where they left off, the system handles this seamlessly in the background, making the overall learning experience smoother and more rewarding.

1.6 Report Organization

The report is organized into ten chapters, each covering a distinct aspect of the system design and implementation. These include process modeling through BPMN diagrams, structural design through class diagrams, behavioral modeling through sequence and

state diagrams, interface design through mock UI, and finally the full feature implementation with deployment details. Each chapter reflects the collaborative effort of the group members who contributed to the respective components of this project.

Chapter 2

BPMN Diagram

Presented by: Tasnimzaman Tanmi (2105147)

2.1 Overview

The BPMN diagram models the complete end-to-end business process of the Udemmy Course Management System, capturing the interactions and workflows across all major stakeholders. The diagram is structured into four pools representing the key actors: **Student**, **Instructor**, **Admin**, and the **System**. The System pool is further subdivided into five lanes — **Progress Tracker**, **Review Manager**, **Analytics Generator**, **Content Manager**, and **Course Manager** — each encapsulating a distinct set of backend responsibilities.

The diagram covers the following major process flows:

- Course creation, content upload, and publishing by the Instructor
- Student course browsing, enrollment, and video playback
- Automatic progress tracking and notification by the System
- Review submission, moderation, and admin decision-making
- Analytics generation for both instructors and the system
- Resource and video download management with access control

2.2 Changes Made

Two modifications were made to the BPMN diagram based on feedback received during the presentation:

1. **Removal of the 20% completion check for review submission** — In the original diagram, a gateway was included that checked whether a student had completed at least 20% of a course before being allowed to submit a review. This conditional check has been removed; students can now submit a review without any completion threshold requirement.
2. **Simplification of download and upload activity sequences** — The original diagram contained redundant activity boxes for both download and upload flows, specifically the three-step sequence of *Start Download* → *Wait for Download* → *Complete Download* (and its upload equivalent). These have been consolidated into a single **Complete Download** (and **Complete Upload**) activity box, eliminating unnecessary redundancy and improving diagram clarity.

2.3 Diagram

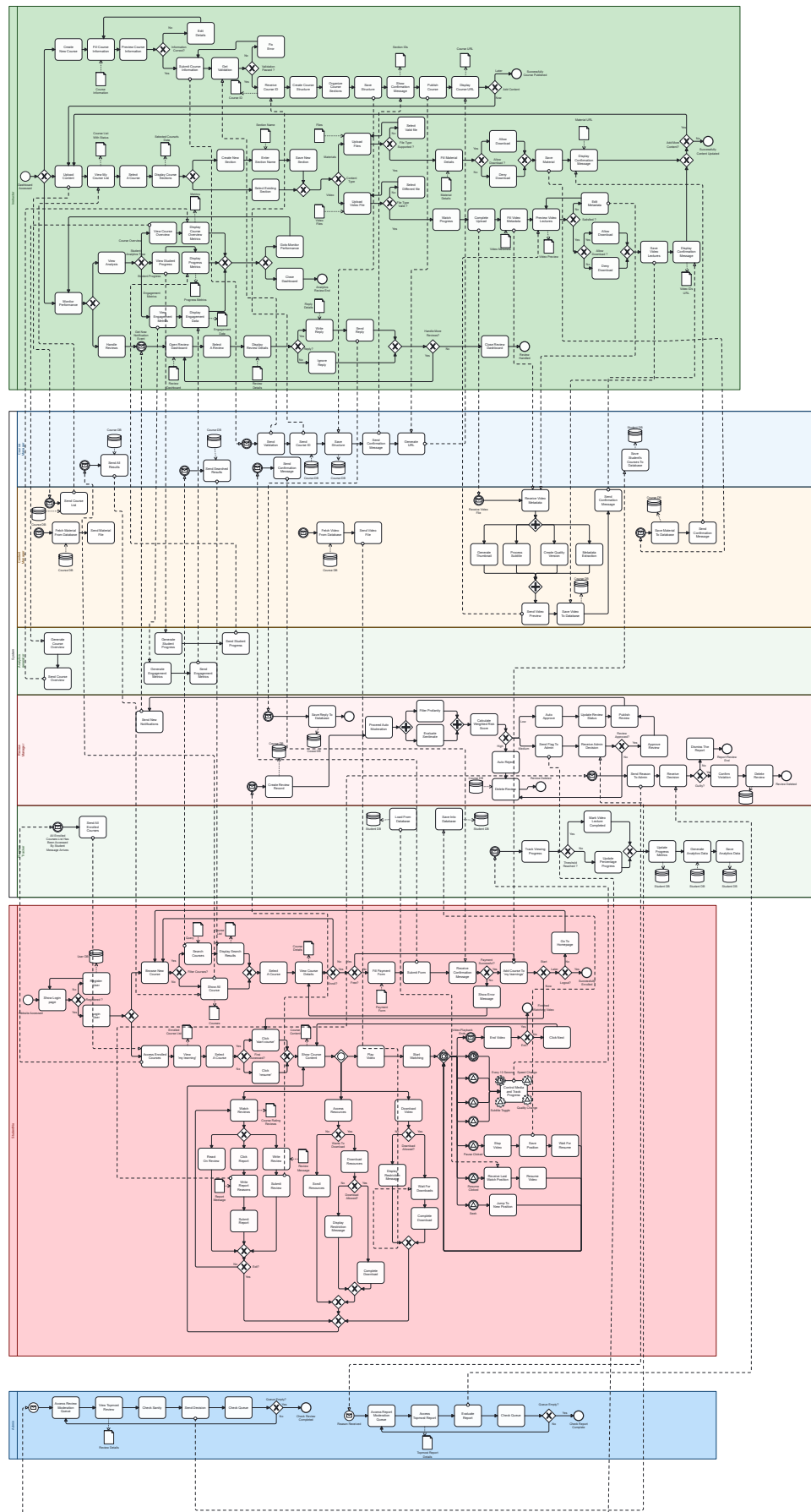


Figure 2.1: BPMN Diagram
6

Chapter 3

Usecase Diagram

Presented by : Nusrat Jahan Tamanna (2105140)

3.1 Overview

This section presents the Use Case Diagram of the online learning platform system Udemey. The diagram shows the interaction between three main actors: Instructor, Student, and Admin. Instructors create and manage courses, upload learning materials, and monitor student engagement through analytics. Students can browse available courses, enroll in courses, watch video lectures, download resources, and provide ratings and reviews. The Admin manages system activities by evaluating reports and approving or rejecting reviews. Overall, the use case diagram illustrates how different users interact with the system and its core functionalities.

3.2 Changes Made

No feedback was provided during the presentation. All diagrams remain unchanged from the original submission.

3.3 Diagram

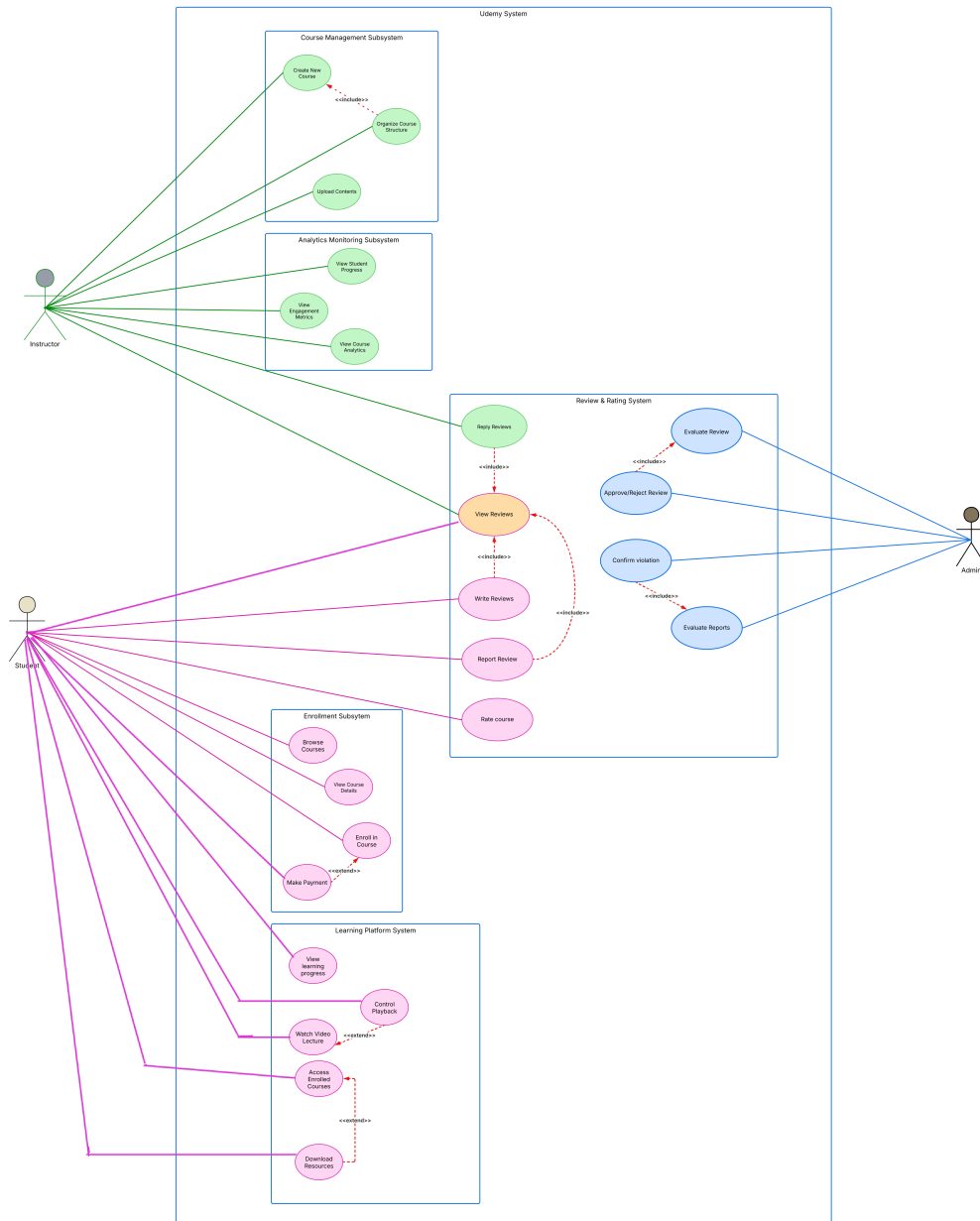


Figure 3.1: Use Case Diagram of the Online Learning Platform Udemy

Chapter 4

Class Diagram

Presented by : Nusrat Jahan Tamanna (2105140)

4.1 Overview

The diagram represents the class structure of an online learning platform Udemy. It shows how different modules of the system interact to manage courses and user activities. The Authentication and User Management System handles user login, access, and account control. The Course Browse and Enrollment module allows users to search, view course lists, check course details, and enroll in courses. Interface classes such as CourseListUI, CourseDetailsUI, and EnrolledCoursesUI display course information and manage user interactions. The system also includes a Course Management System for organizing course content and structure. A Video Playback and Progress System tracks learning progress while students watch course materials. The Review and Rating System allows users to provide feedback on courses. Additionally, an Analytics and Dashboard module helps monitor user activity and course performance. Overall, the diagram illustrates how multiple components work together to deliver a complete online learning experience.

4.2 Changes Made

We updated the class diagram based on the teacher's feedback. Previously, the Admin class was empty in Authentication and User management system, so we added relevant attributes and operations to make it functional within the system. The Admin

class now includes a `moderationQueue` attribute and methods such as `approveReview()`, `rejectReview()`, and `viewModerationQueue()` to handle review moderation tasks.

4.3 Diagrams

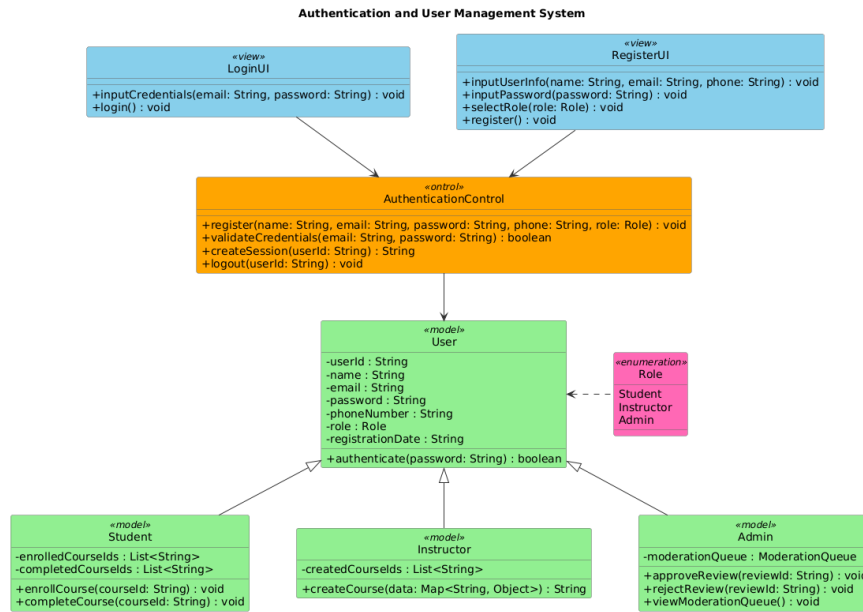


Figure 4.1: Authentication and User Management System

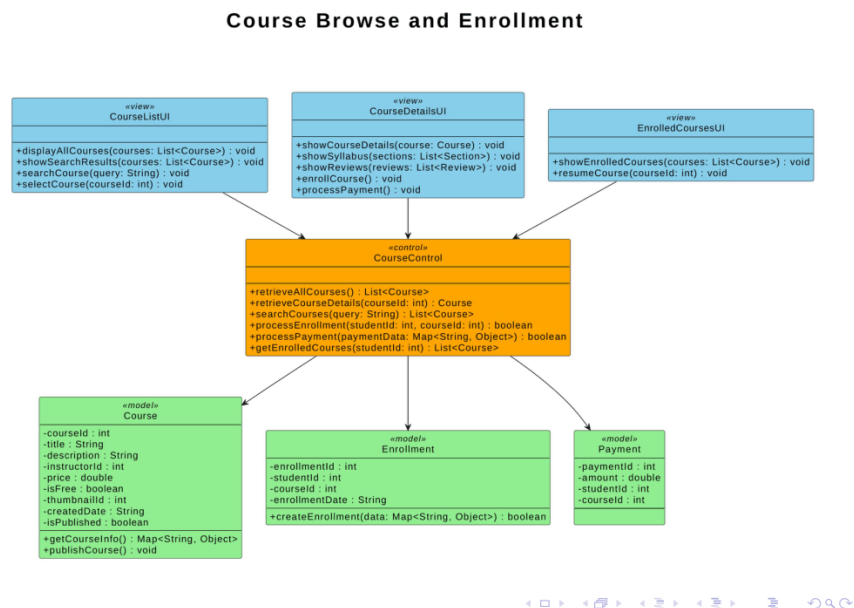


Figure 4.2: Course Browse and Enrollment Diagram

Course Management System

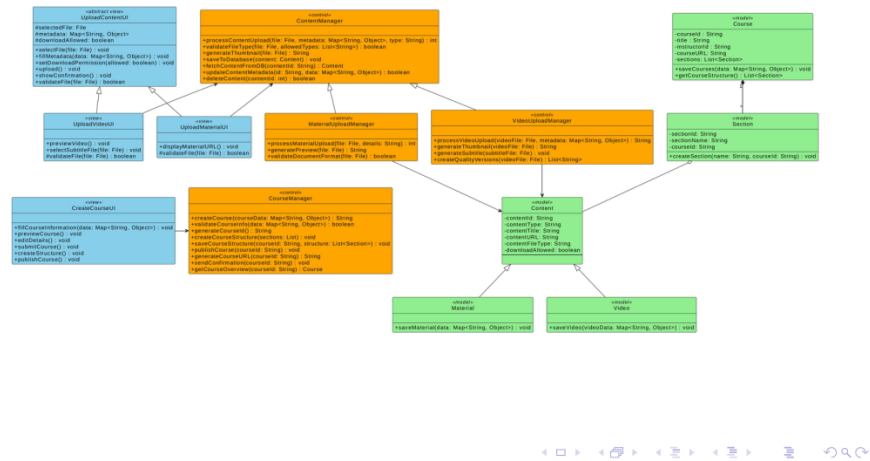


Figure 4.3: Course Management System

Video Playback & Progress System



Figure 4.4: Video Playback and Progress Tracking

Review and Rating System



Figure 4.5: Review and Rating System

Analytics & Dashboard

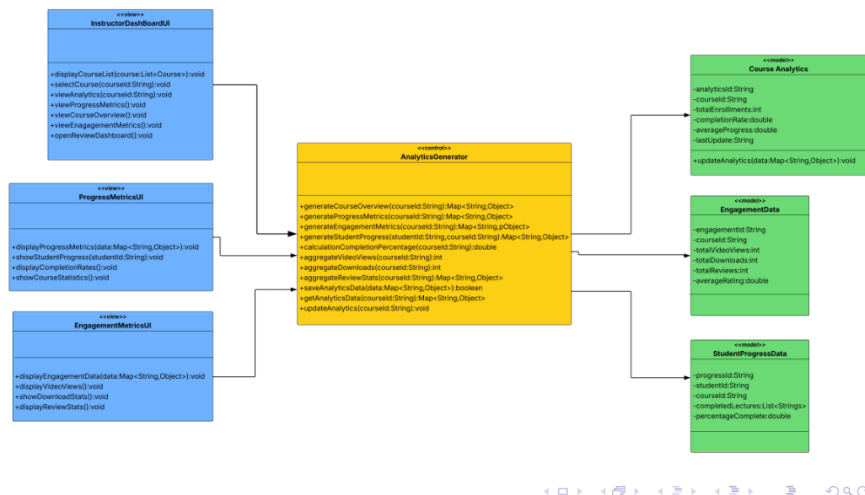


Figure 4.6: Analytics and Dashboard

Chapter 5

Mock UI

Presented by: Shatabdi Dutta Chowdhury (2105123)

5.1 Overview

This chapter presents the high-fidelity Mock UI for the Udemmy Course Management System, designed extensively using Figma. The primary goal of these mockups is to visualize the complete end-to-end user experience for both students and instructors. It translates our project's core requirements—such as video-based playback, automatic progress tracking, resource downloading, and interactive community reviews—into an intuitive, interactive frontend interface. By mapping out every state from public access and authentication to course creation and performance analytics, these diagrams serve as the definitive blueprint for the system's implementation.

5.2 Changes Made

During the project presentation, the evaluation panel identified a few usability and layout issues. Based on this valuable feedback, the following bugs were successfully addressed and fixed in the final prototype presented below:

- **Bug 1 (Course Card Design):** In the initial prototype, there was a layout inconsistency and alignment bug in the course cards displayed on the category dashboard. This has been corrected to ensure clear typography, and a polished visual hierarchy across all horizontal scrolling lists.

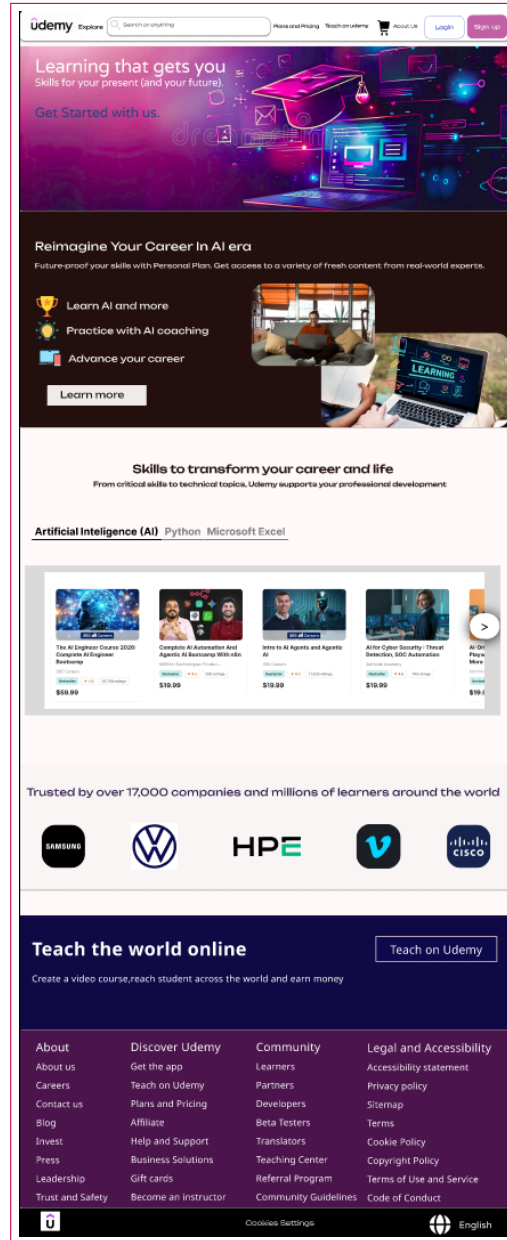
- **Bug 2 (Instructor 'Student' Switch):** On the instructor dashboard, the control to switch back to a student profile was originally rendered as plain text, making it non-intuitive and hard to recognize as an interactive element. It has now been redesigned as a distinct, recognizable button to improve user navigation.

5.3 Ultimate Diagram Section

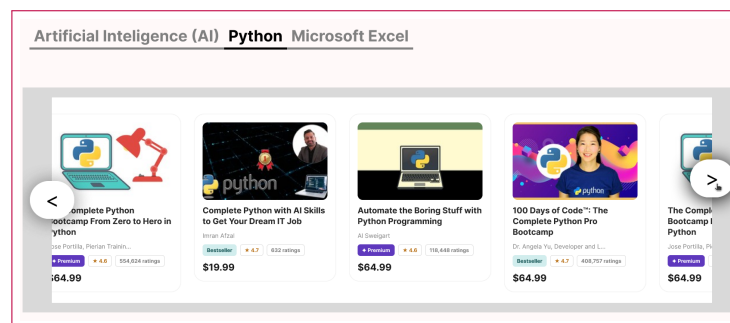
The following subsections meticulously detail the complete user flow across the platform, incorporating all 77 interactive Figma screens developed for this module.

5.3.1 Public Access and Authentication Flow

When a public user lands on the platform without being authenticated, they are greeted by the homepage, which provides a general overview of available courses. To access content, users can either register a new account or log in to an existing one.

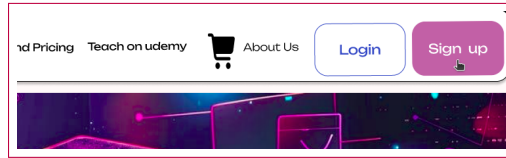


(a) Public Homepage without authentication

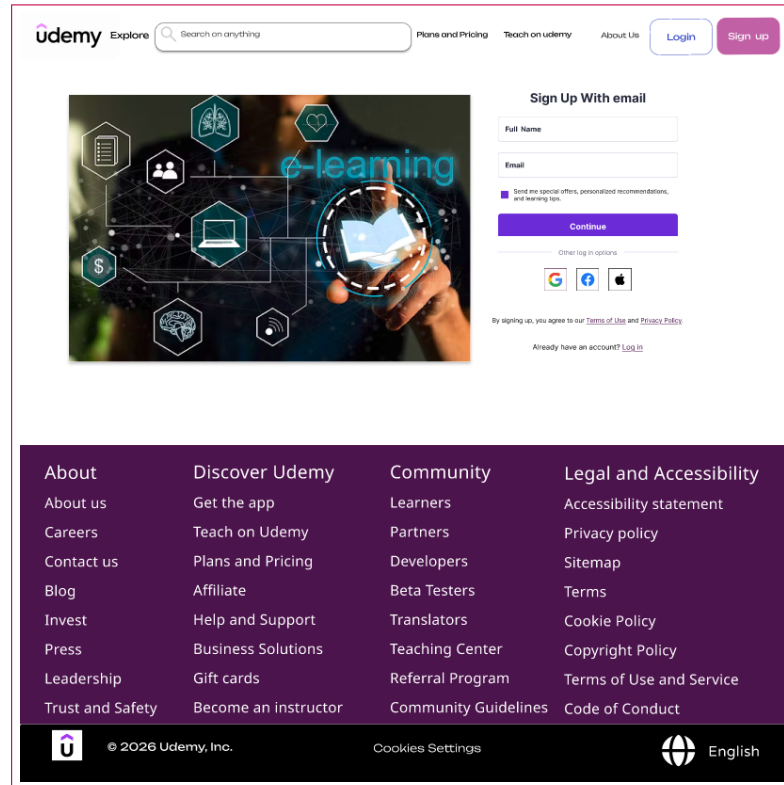


(b) Course list overview visible to the public

Figure 5.1: Public user landing experience



(a) Clicking the Sign Up option



(b) Submitting registration details

Sign Up With email

☒ Send me special offers, personalized recommendations, and learning tips.

Other log in options

By signing up, you agree to our [Terms of Use](#) and [Privacy Policy](#).

Already have an account? [Log in](#)

(c) Sign Up Form Page

Figure 5.2: New User Registration Flow

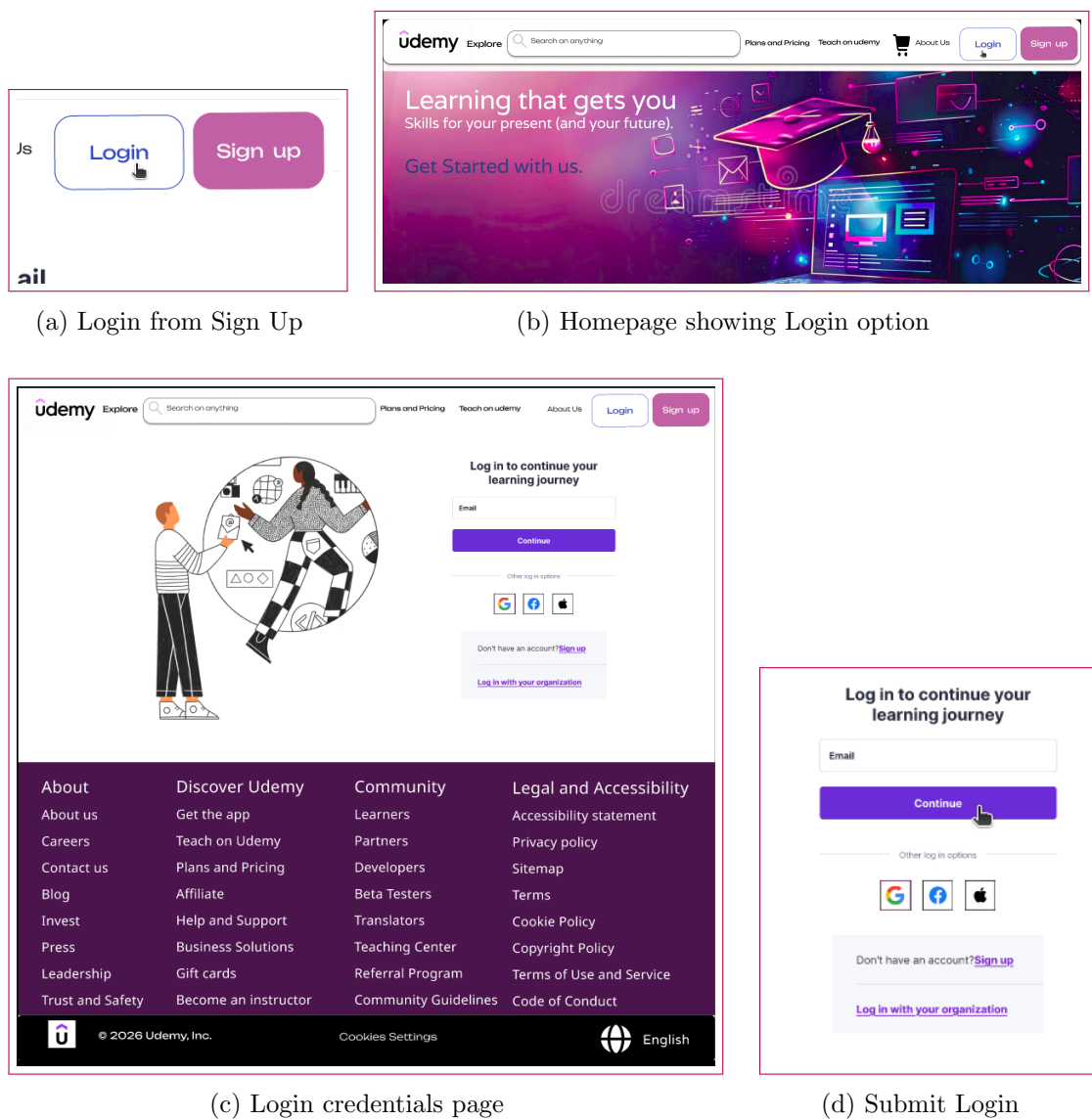
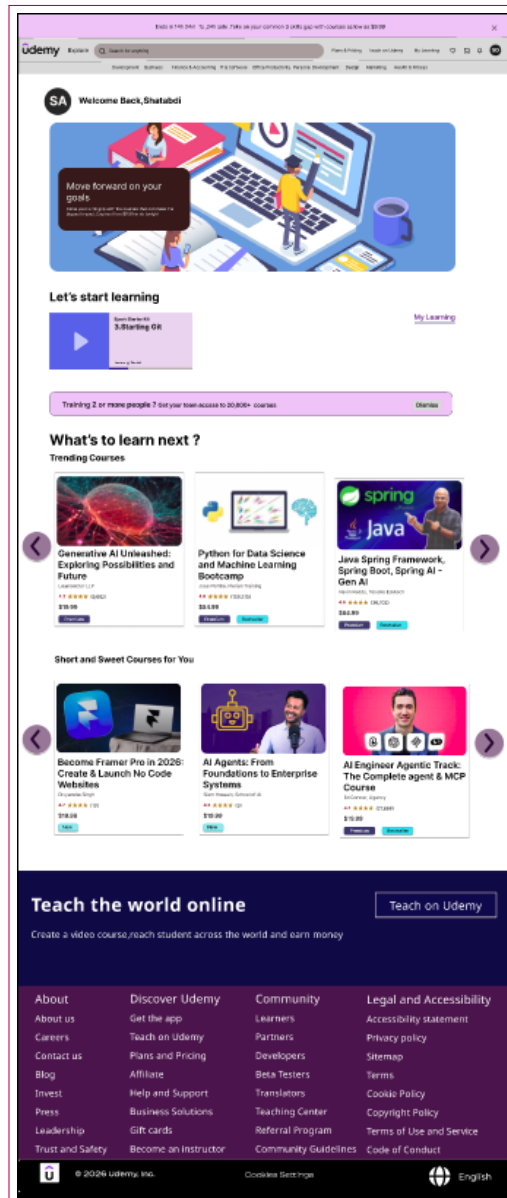


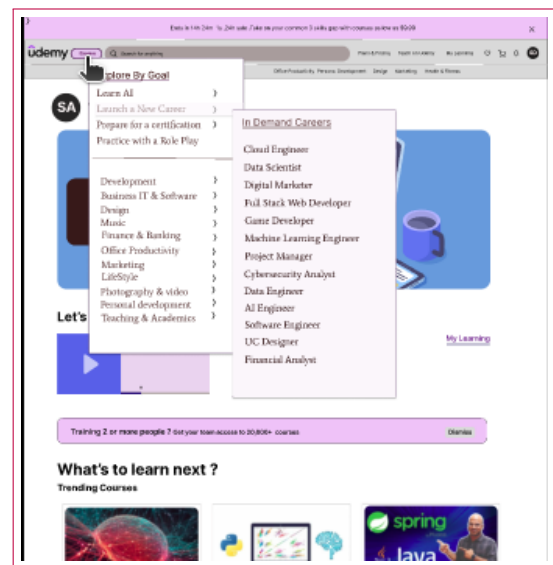
Figure 5.3: Existing User Login Flow

5.3.2 Student Dashboard and Course Discovery

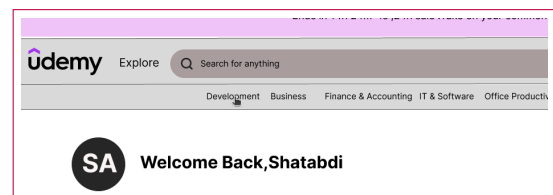
Upon successful authentication, the user is redirected to their personalized Student Dashboard. From here, they can navigate through categories to discover tailored courses and trending topics.



(a) Authenticated Student Dashboard

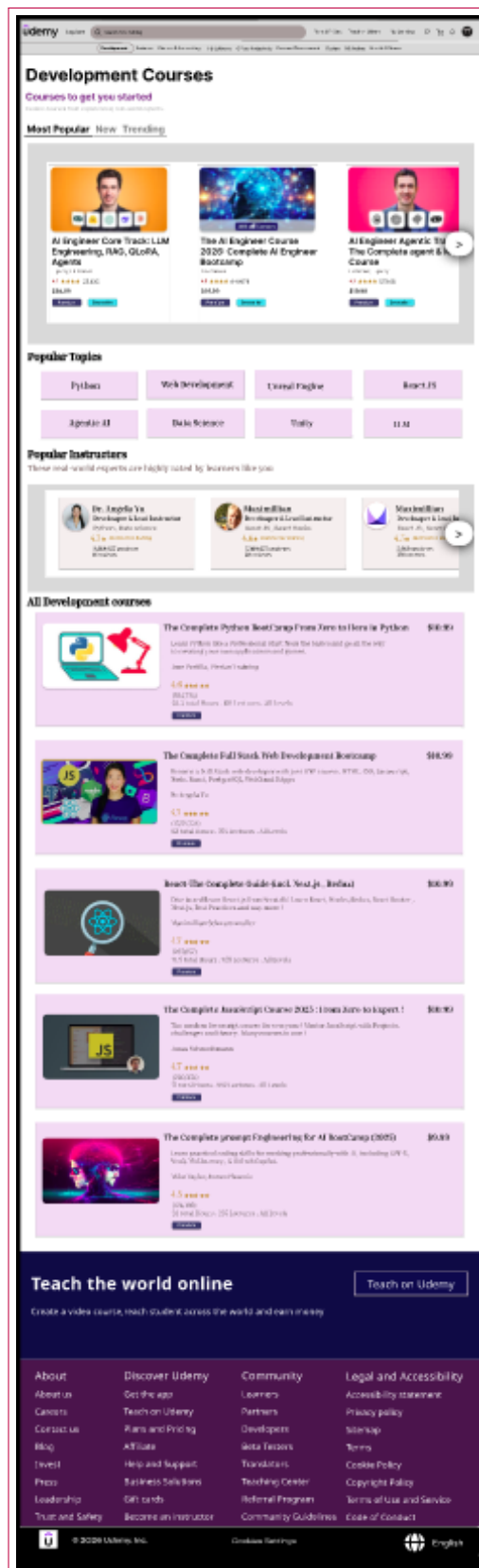


(b) Explore dropdown menu

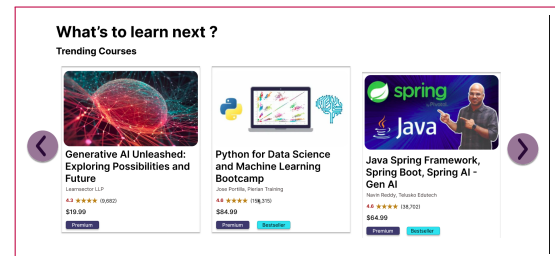


(c) Selecting the Development category

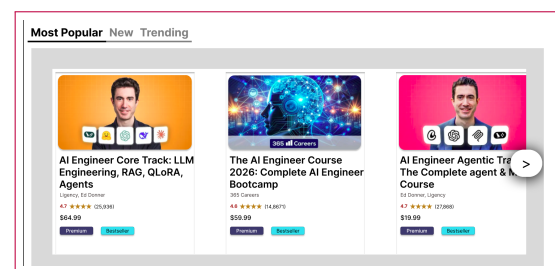
Figure 5.4: Navigation and Category Selection



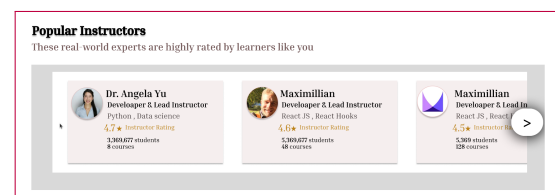
(a) The Development Category Dashboard



(b) Corrected horizontal scrolling list of trending courses (Bug 1 Fixed)



(c) Horizontal list showing "Most Popular" courses



(d) Popular Instructors highlighted within the category

Figure 5.5: Category Dashboard and Discovery Lists

5.3.3 Course Details and Enrollment

When a student finds a course of interest, they navigate to the comprehensive Course Details page. Here, they can review the curriculum, check authentic feedback, and initiate the purchasing flow.

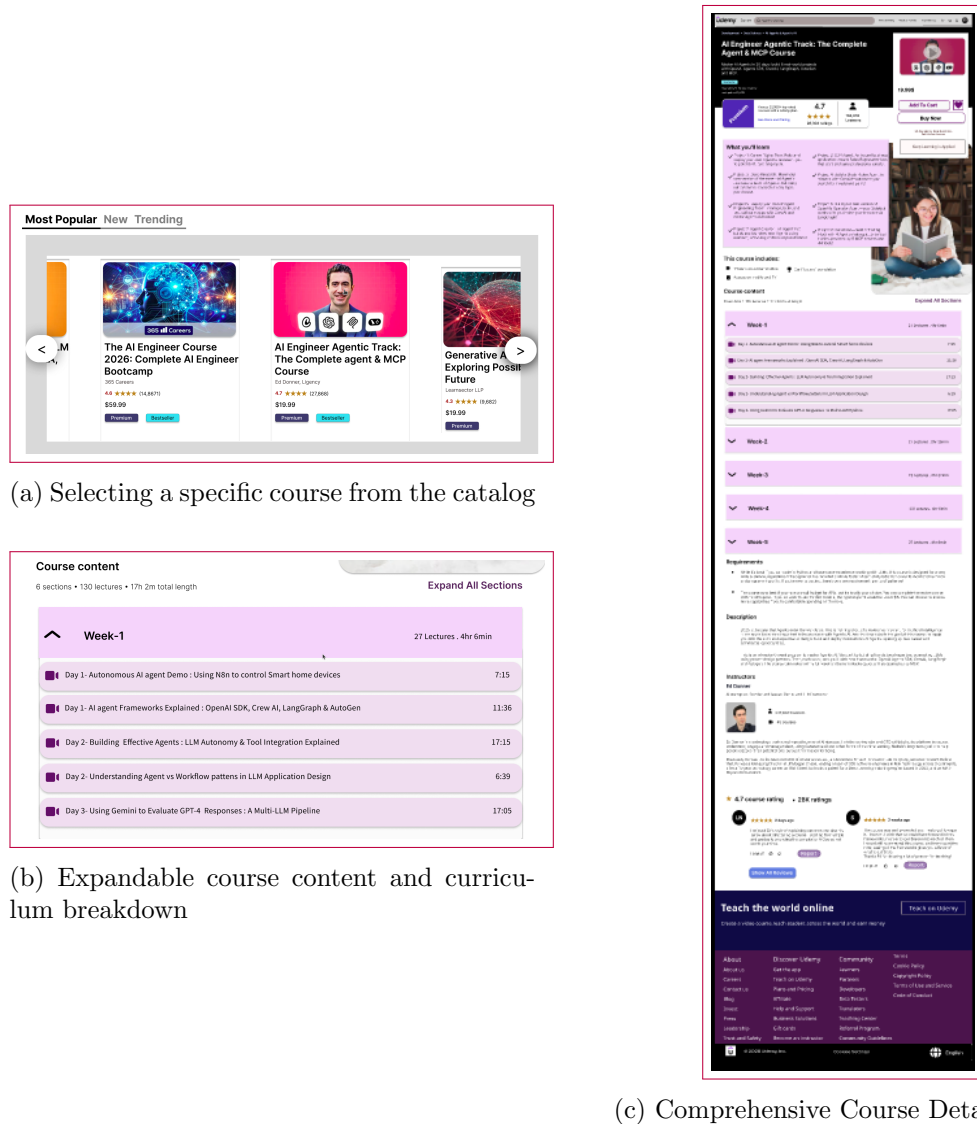
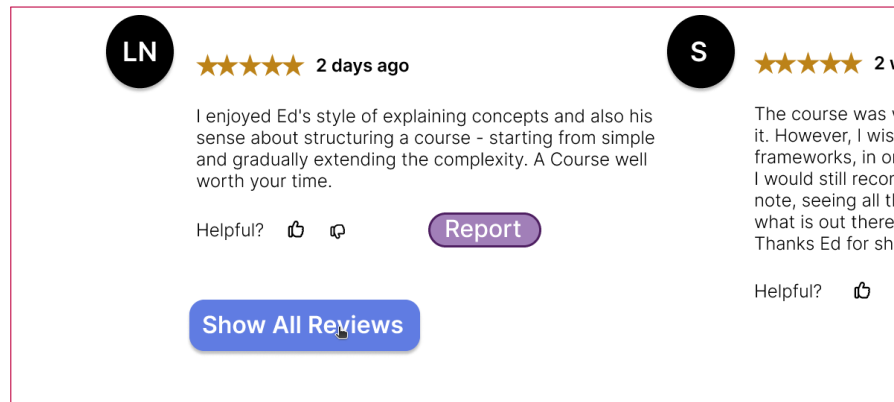
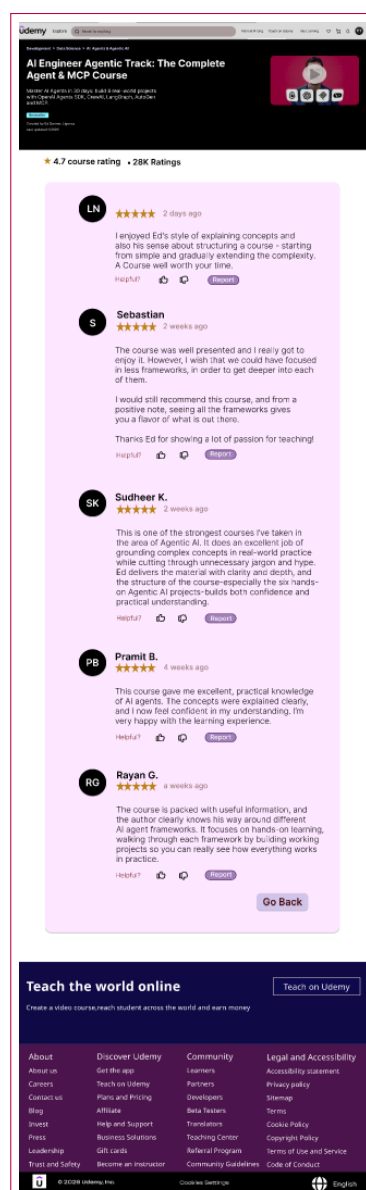


Figure 5.6: Course Details and Curriculum

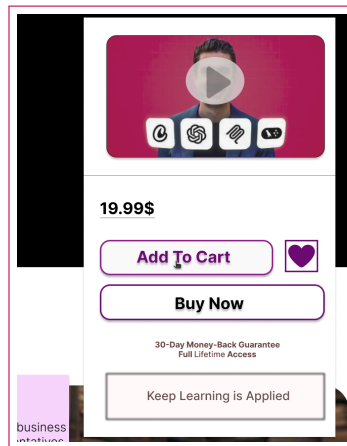


(a) Course review section and option to view all feedback

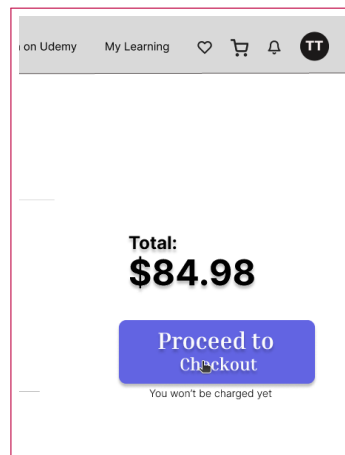


(b) Enrolled students can actively like, dislike, or report reviews

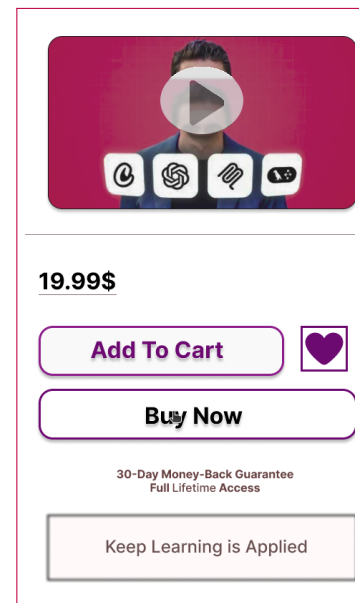
Figure 5.7: Rating and Review Interactions



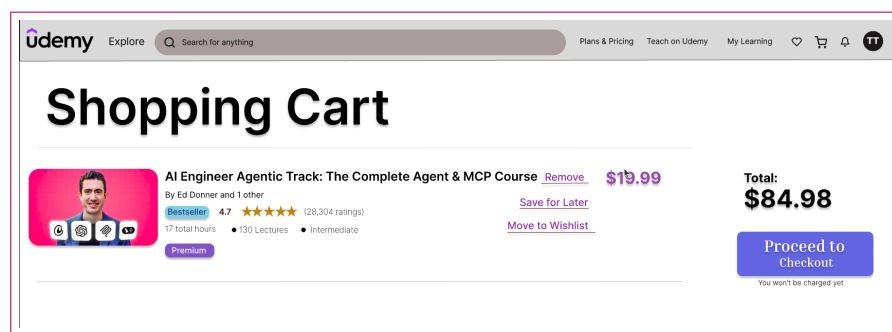
(a) Add to Cart button



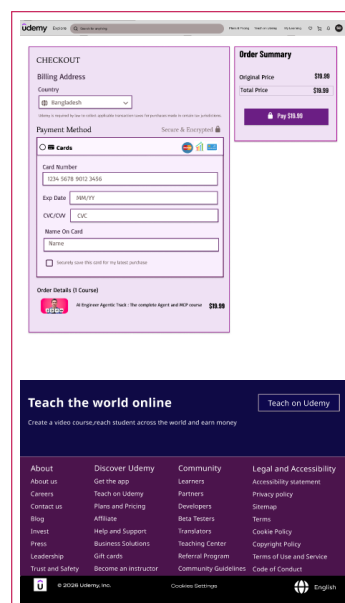
(b) Proceed to Checkout



(c) Direct Buy Now option



(d) Shopping Cart overview

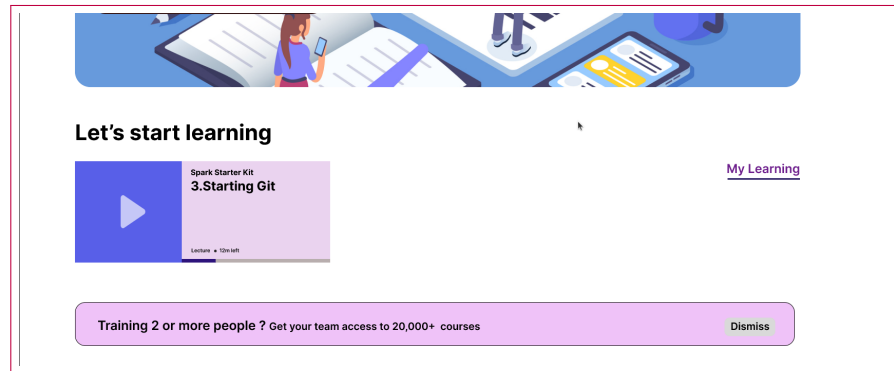


(e) The final Checkout and Payment form

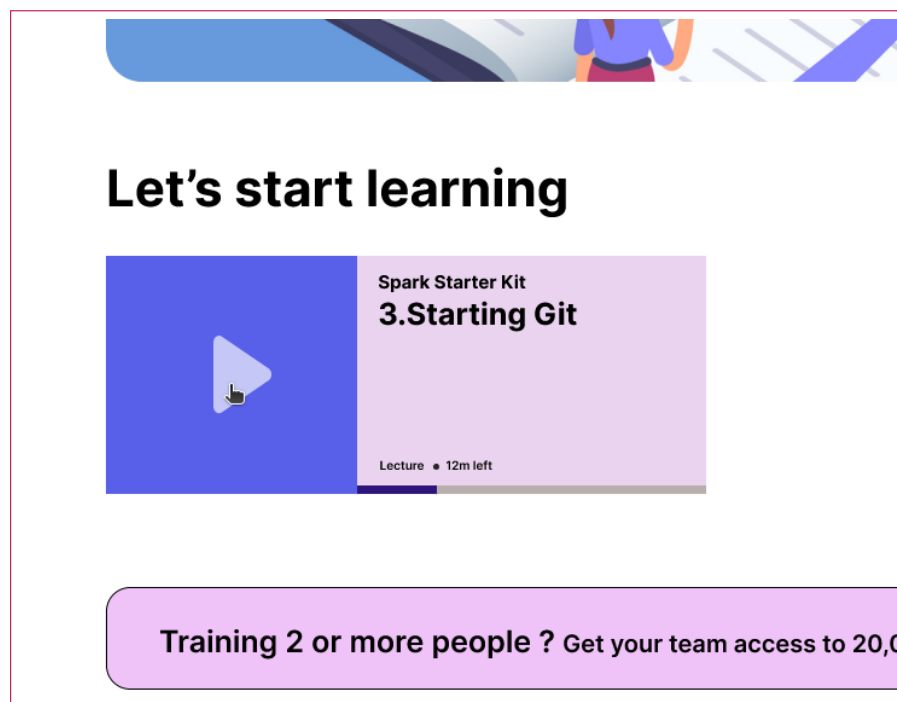
Figure 5.8: Checkout and Purchasing Flow

5.3.4 Video Playback, Progress Tracking, and My Learning

Our assigned core feature—Automatic Progress Tracking—is deeply integrated into the video playback and learning management dashboards, ensuring students can easily monitor and resume their journey.

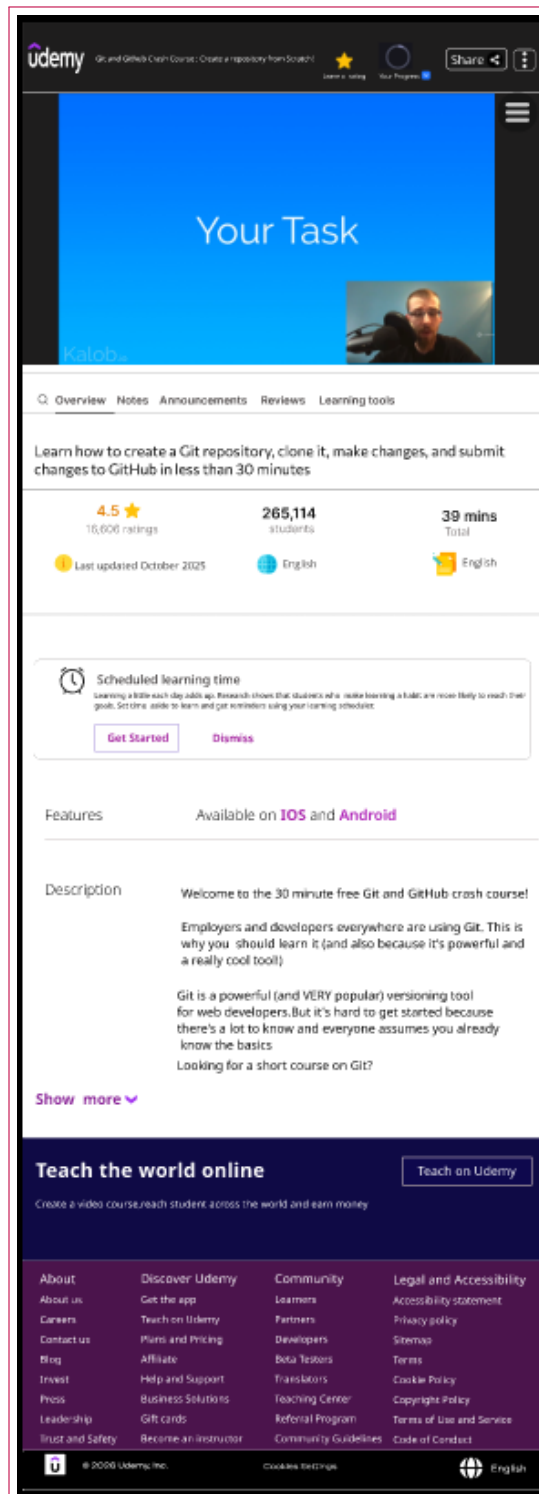


(a) Dashboard widget to quickly resume recent videos

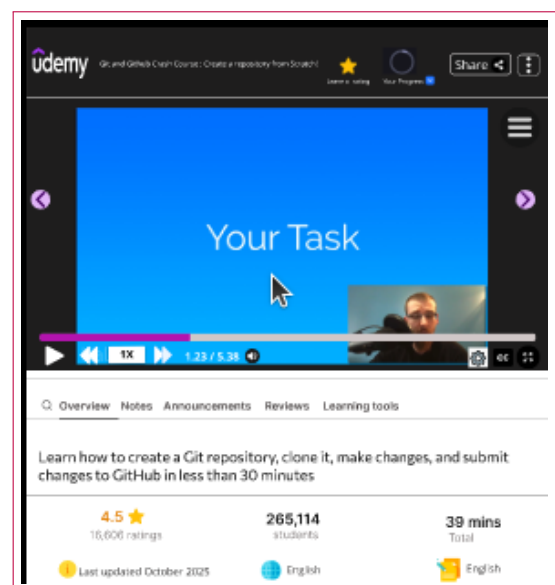


(b) The active 'Let's Start Learning' player widget

Figure 5.9: Resuming Course Content

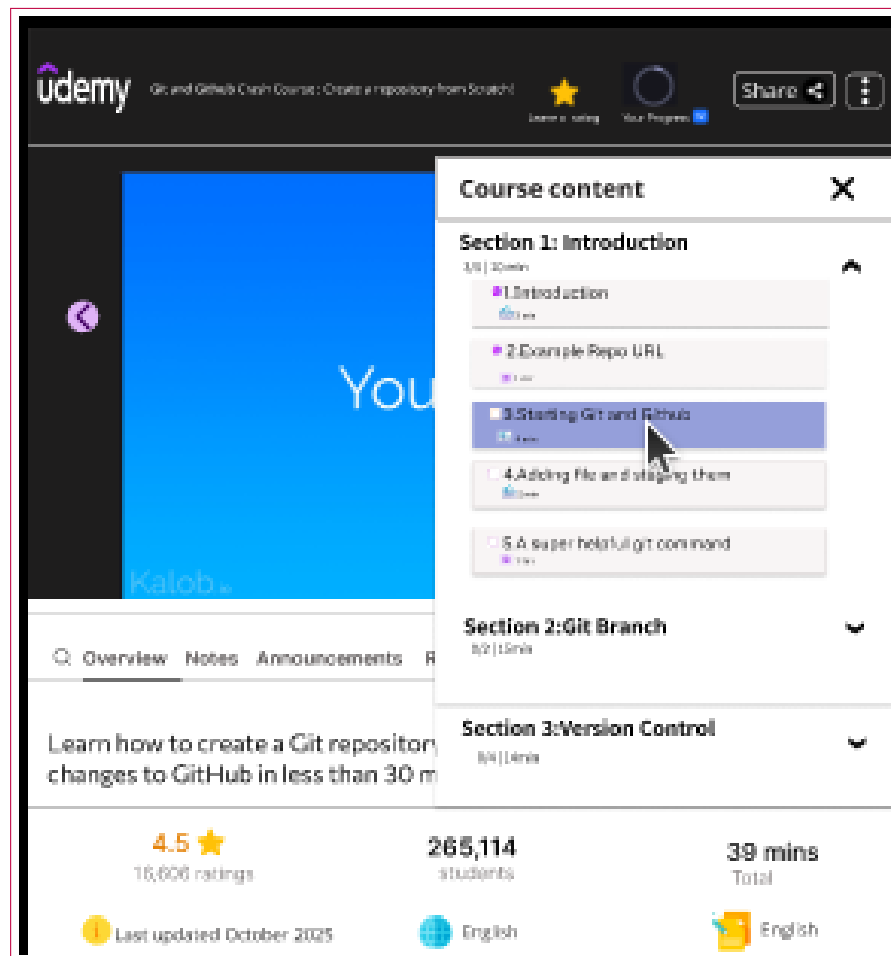


(a) Video playback landing page

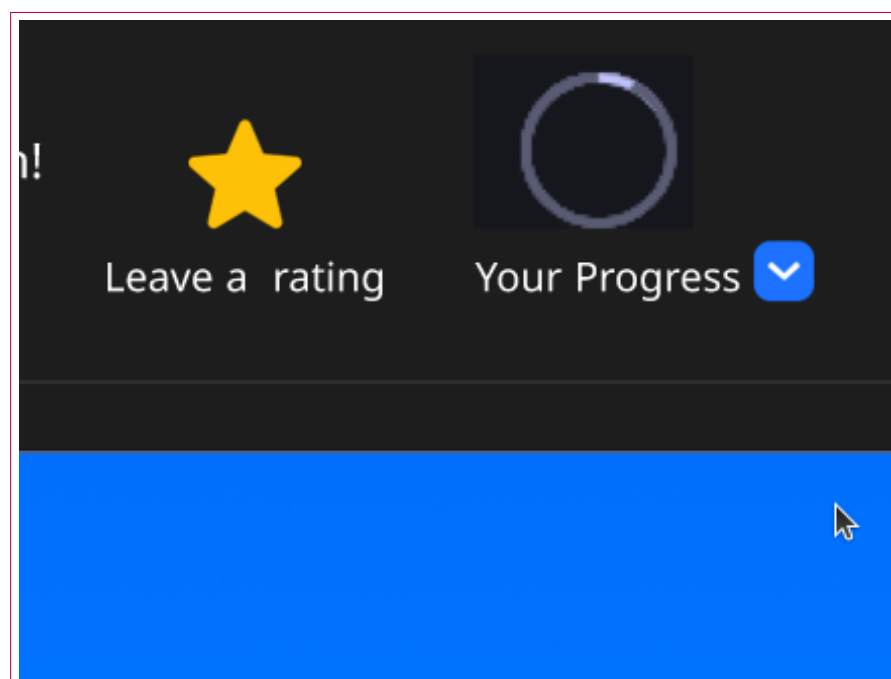


(b) Interactive video playback controls on hover

Figure 5.10: Video Player Interface

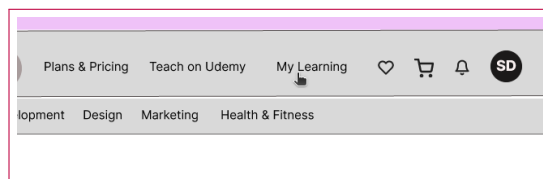


(a) Sidebar menu for course sections

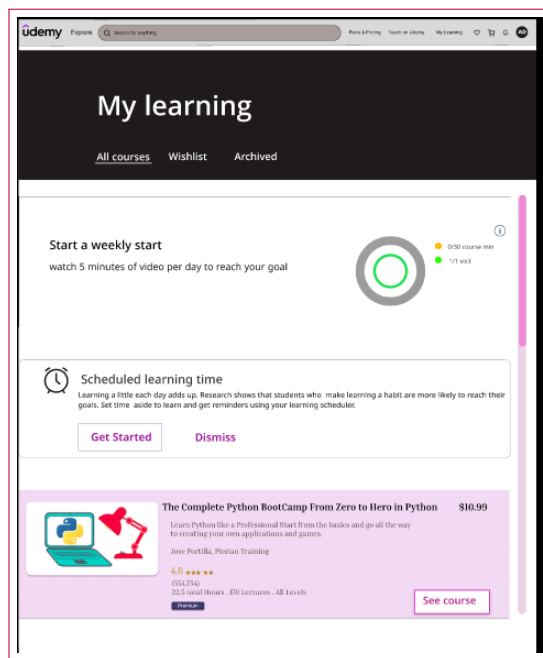


(b) Visual Progress Tracking indicator

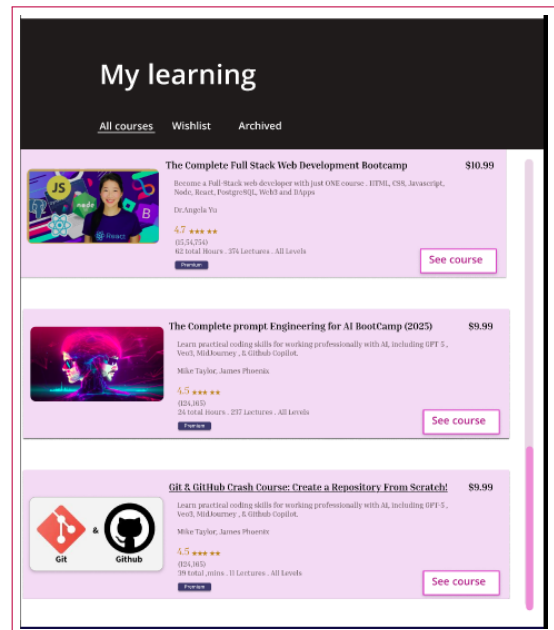
Figure 5.11: Course Navigation and Progress Analytics



(a) Navigating to 'My Learning'

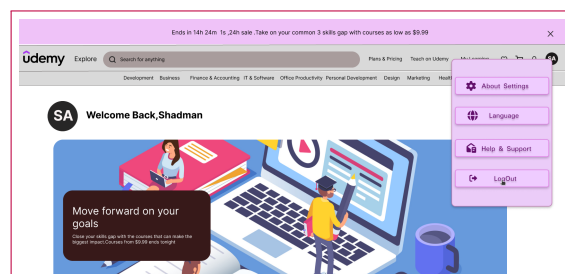


(b) My Learning dashboard showing viewing goals



(c) Vertical list displaying all enrolled courses

Figure 5.12: Managing Enrolled Content and Sessions

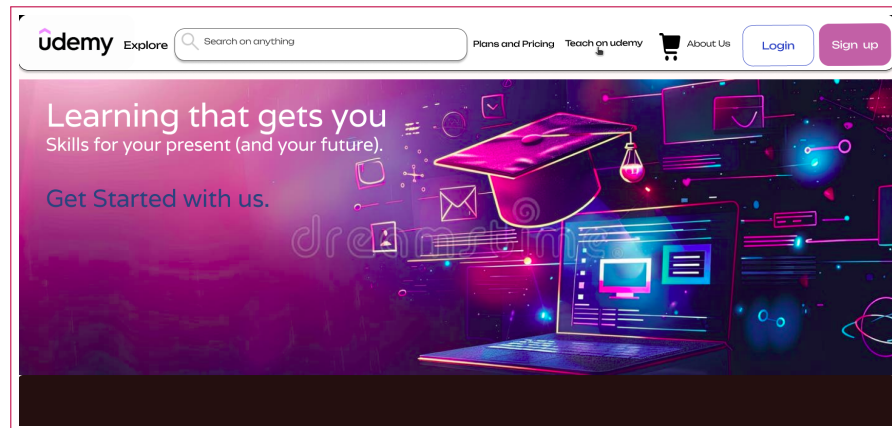


(a) Profile dropdown to safely LogOut

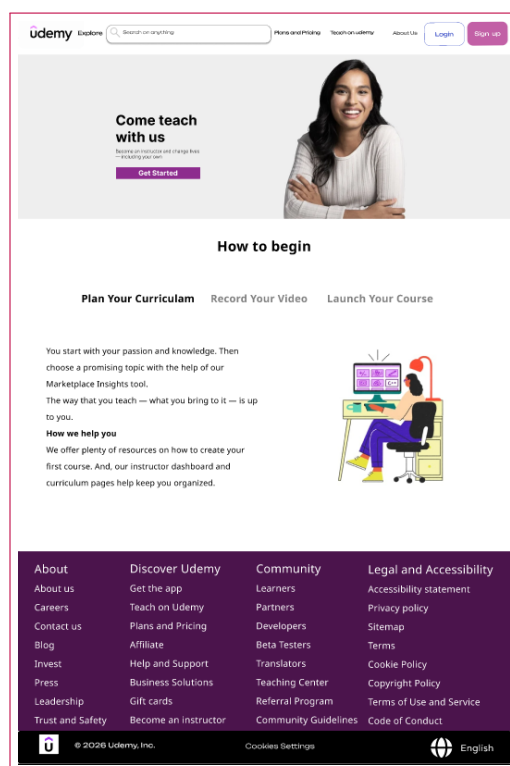
Figure 5.13: Logout from Student site

5.3.5 Instructor Portal and Course Creation

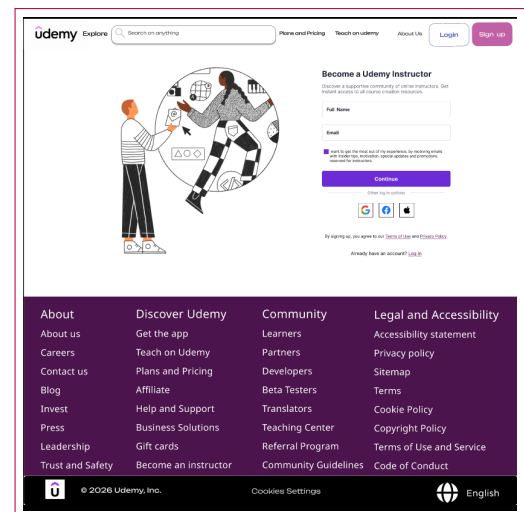
The platform operates on a dual-role system. Any user can navigate to the Instructor Portal to share their expertise. The UI provides a step-by-step wizard for new course initialization.



(a) Homepage link to "Teach on Udeity"

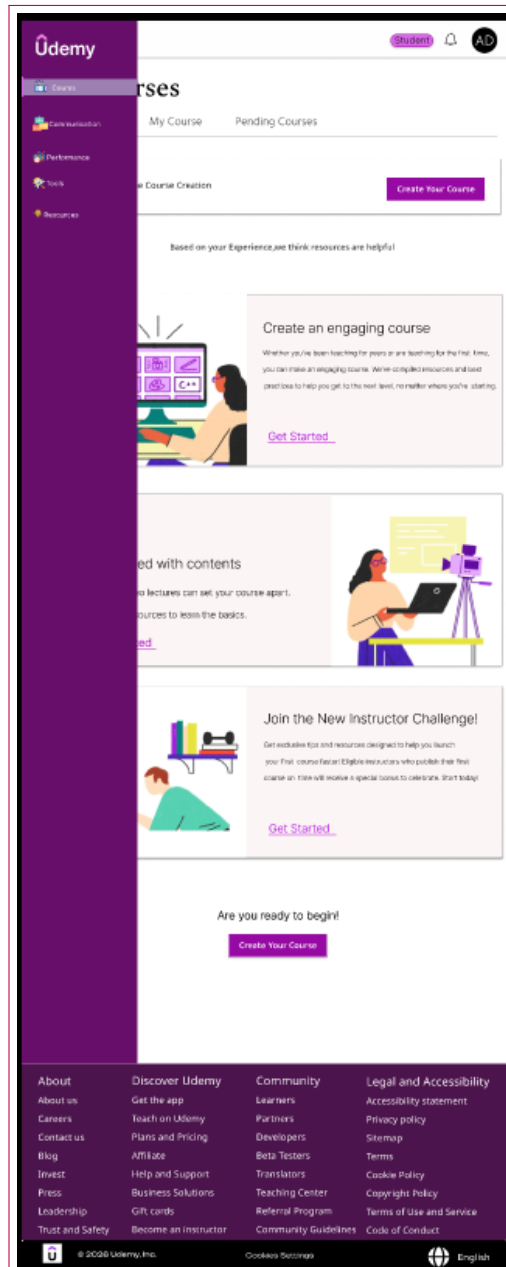


(b) Instructor landing portal

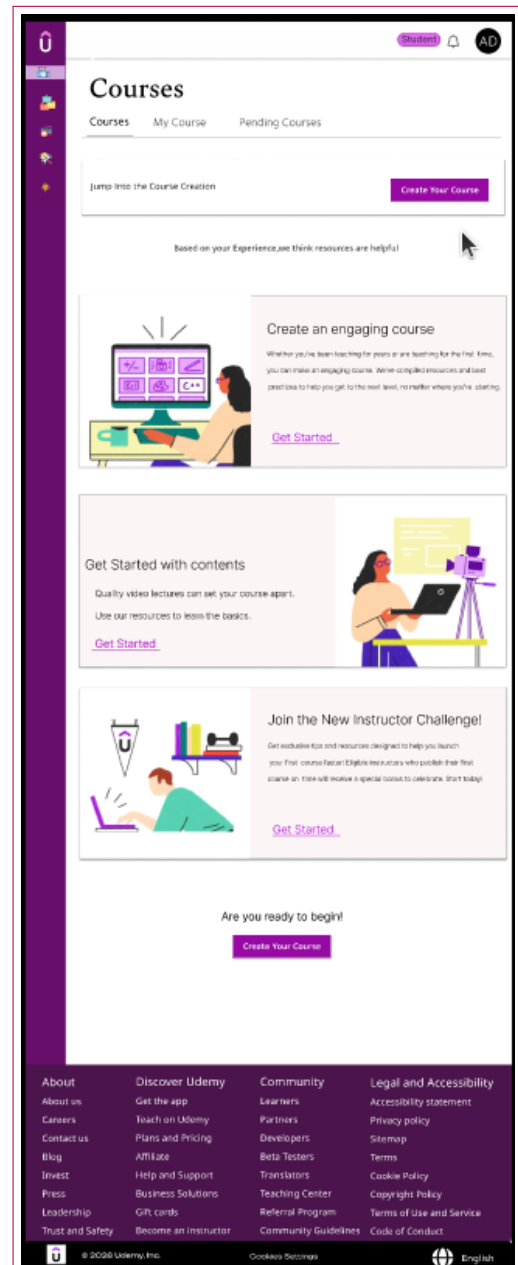


(c) Instructor Registration form

Figure 5.14: Instructor Onboarding Flow

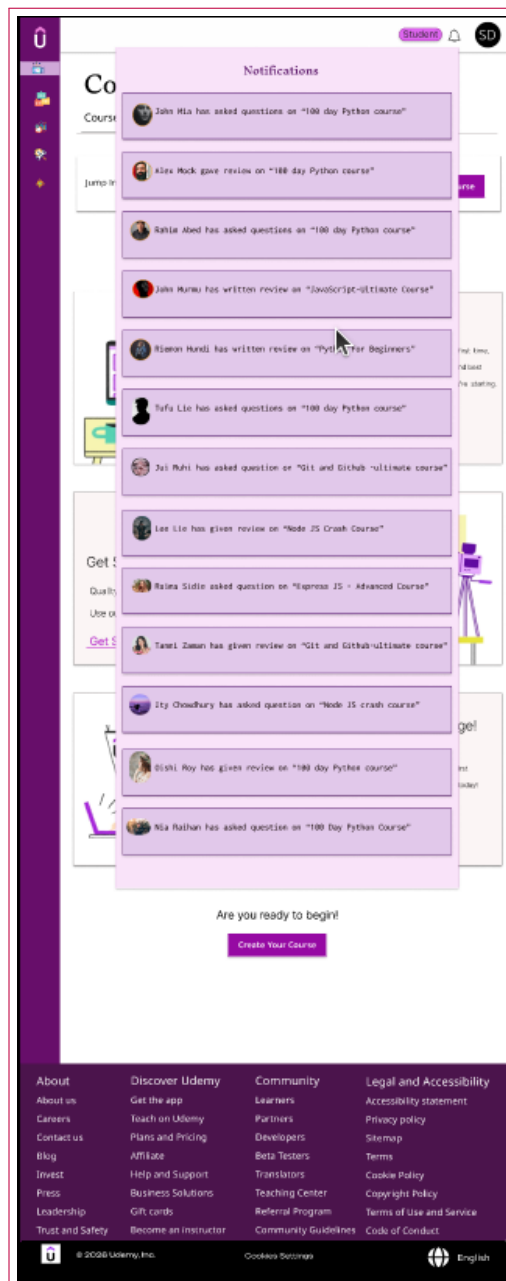


(a) Instructor Dashboard showing the fixed "Student" role-switch button (Bug 2 Fixed)



(b) Navigating to the Courses tab via side-bar

Figure 5.15: Instructor Dashboard Operations

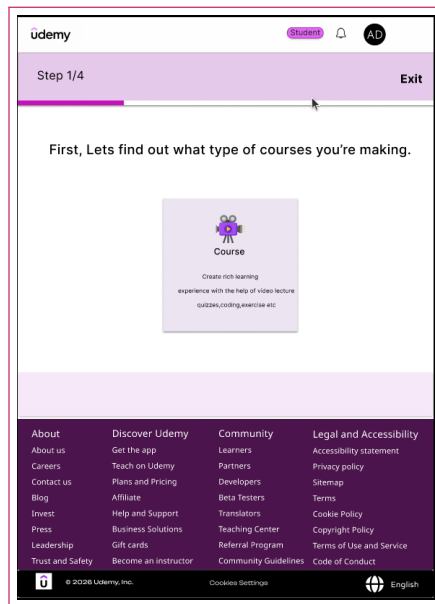


(a) Instructor notifications panel

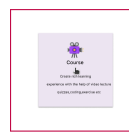


(b) Initiating "Create Your Course"

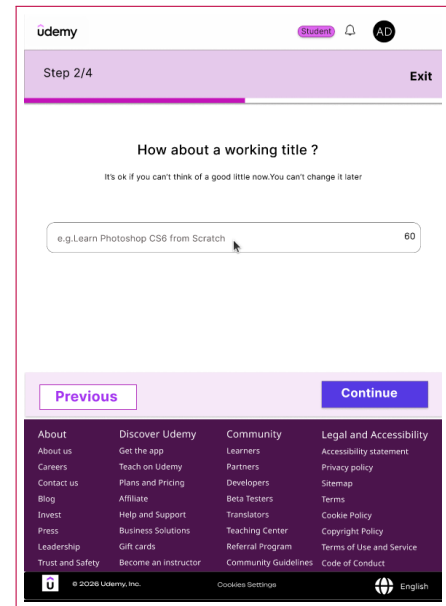
Figure 5.16: Instructor Notification and initiate course creation



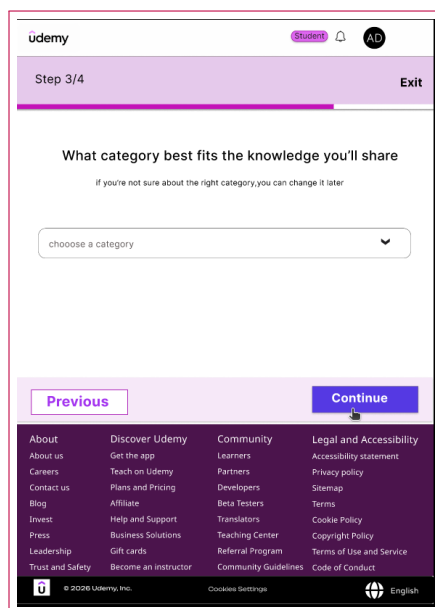
(a) Wizard Step 1: Interface



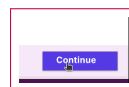
(b) Click course creation



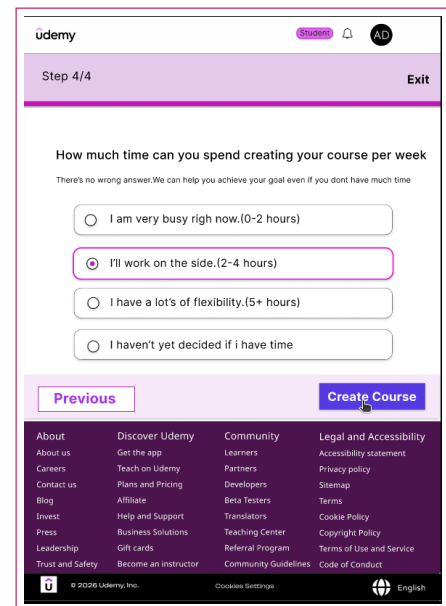
(c) Wizard Step 2: Course Title



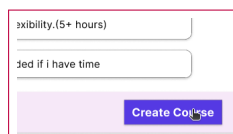
(d) Wizard Step 3: Category



(e) Next Step

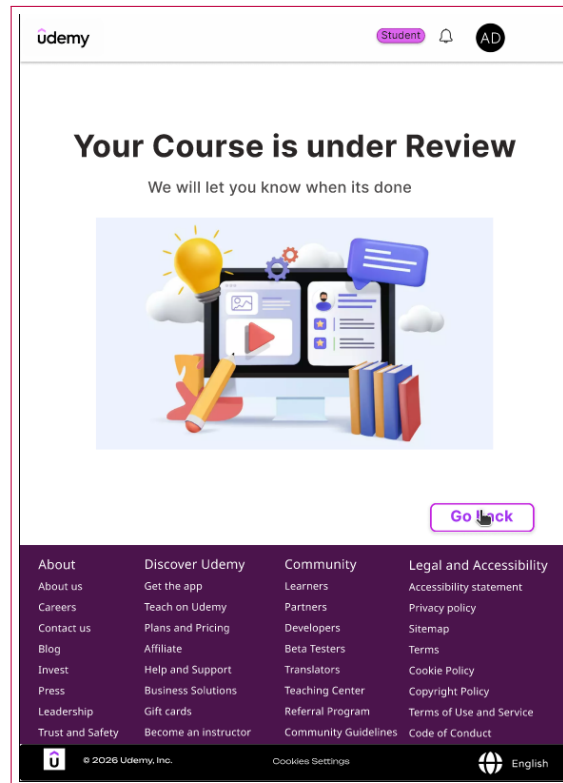


(f) Wizard Step 4: Time Commitment



(g) Create Course

Figure 5.17: The 4-Step Course Creation Wizard

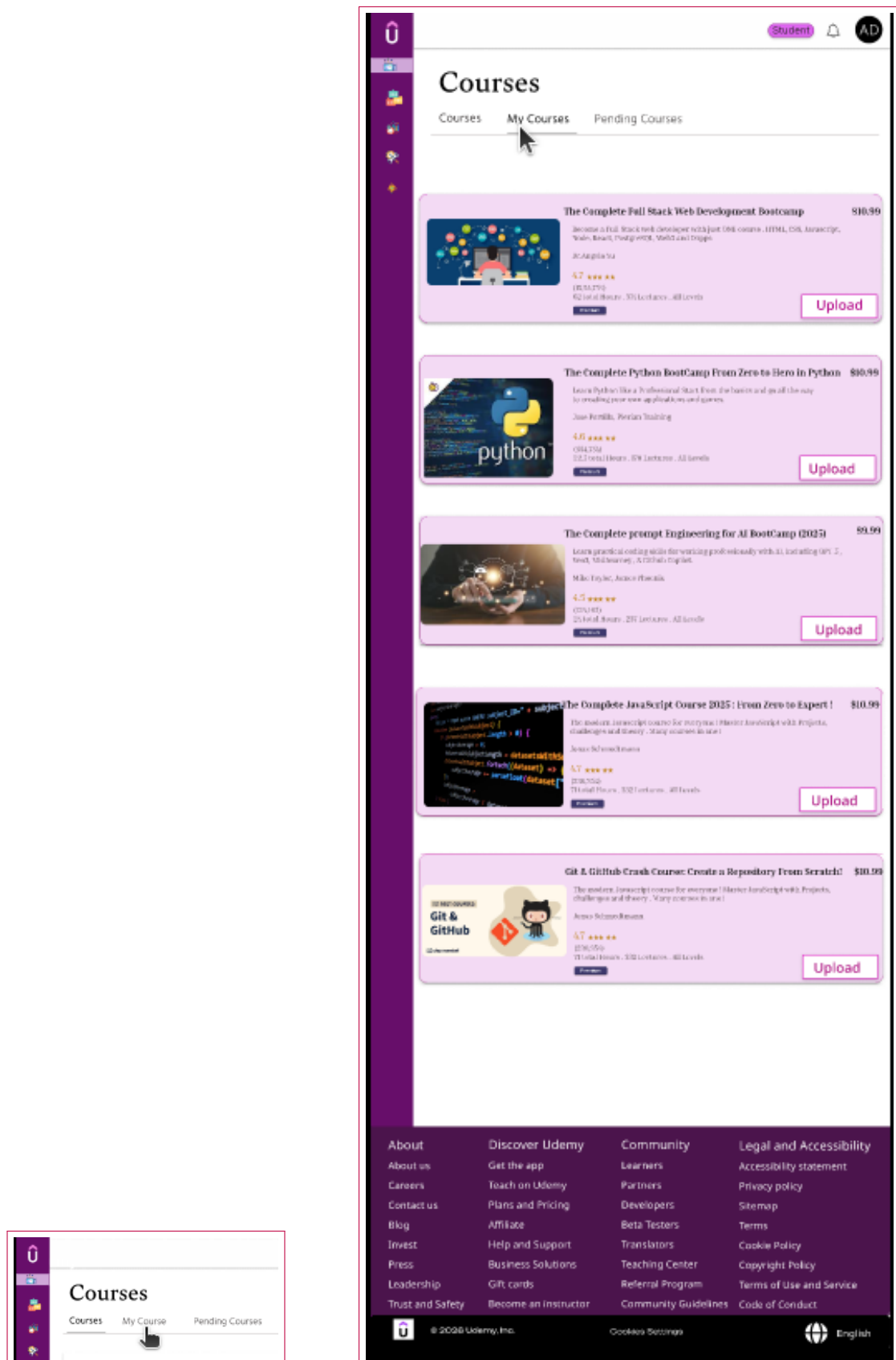


(a) Confirmation: Course is Under Review

Figure 5.18: Pending course

5.3.6 Instructor Content Upload and Moderation

After a framework is built, instructors can view their portfolio, upload dynamic video content, and track the approval status of their courses.



(a) Clicking 'My Course'

(b) Published course list with Upload options

Figure 5.19: Accessing Published Content

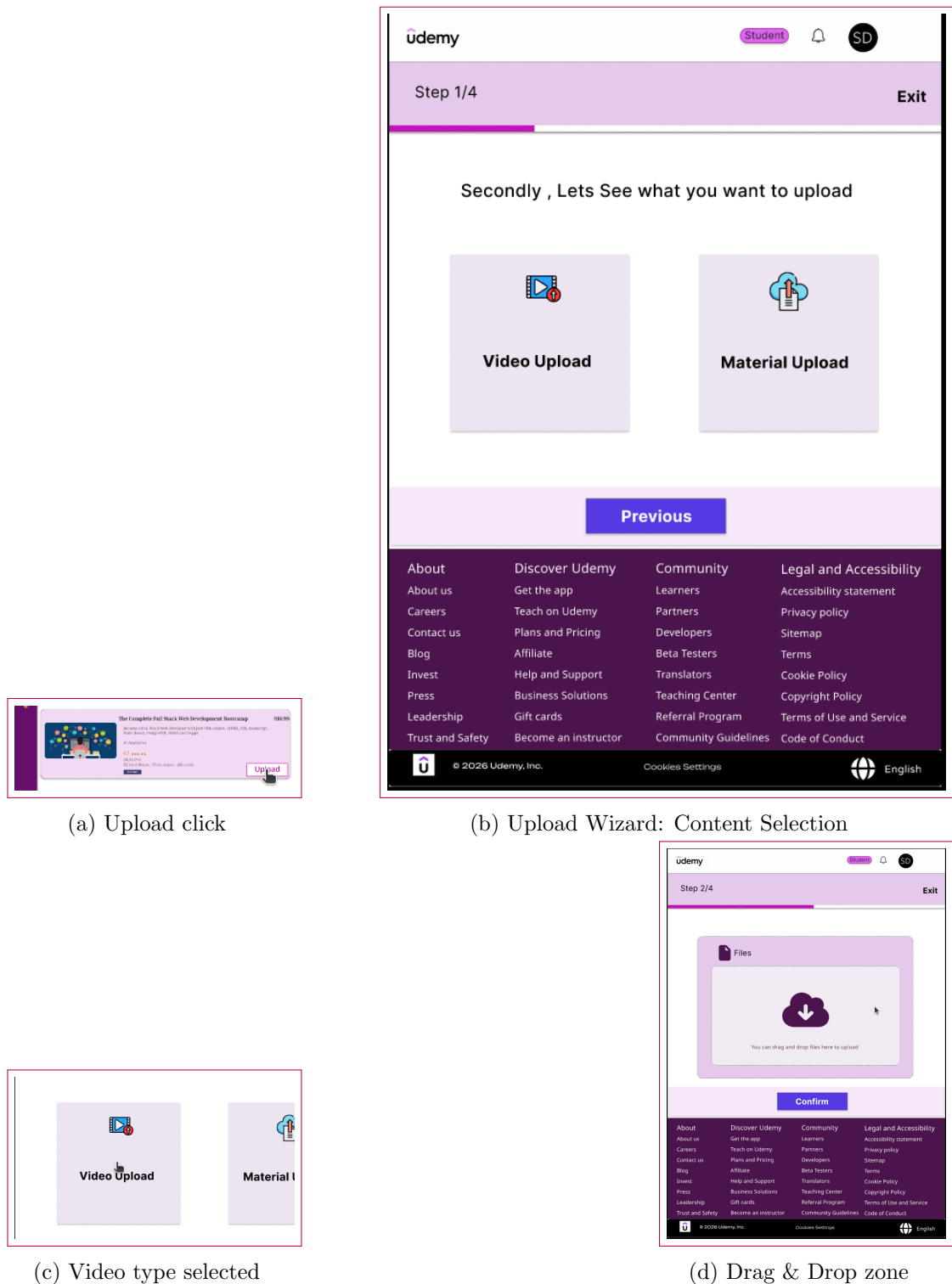


Figure 5.20: Content Upload Process Continue

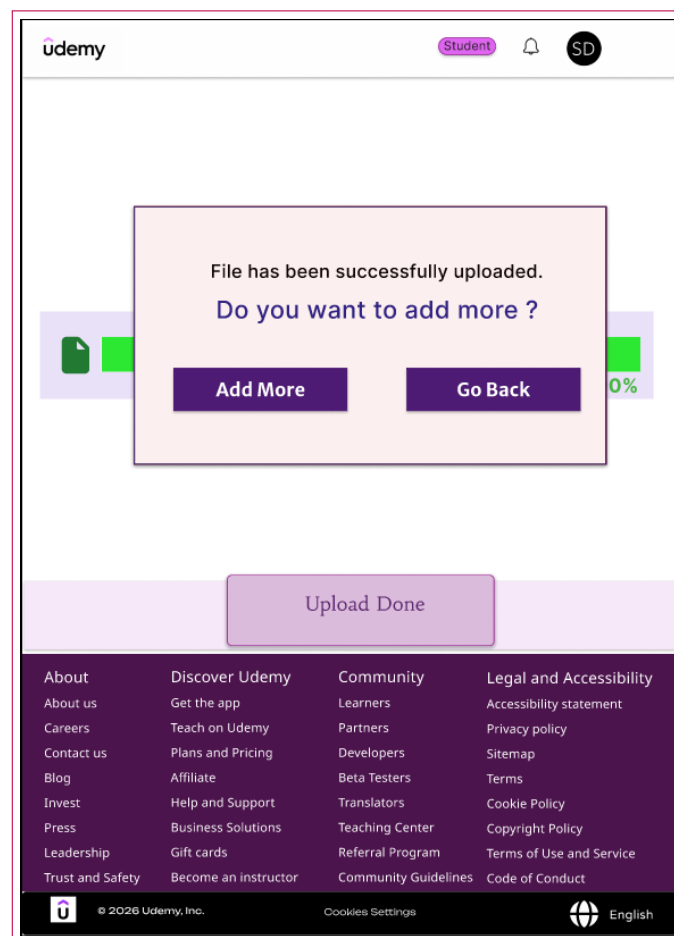
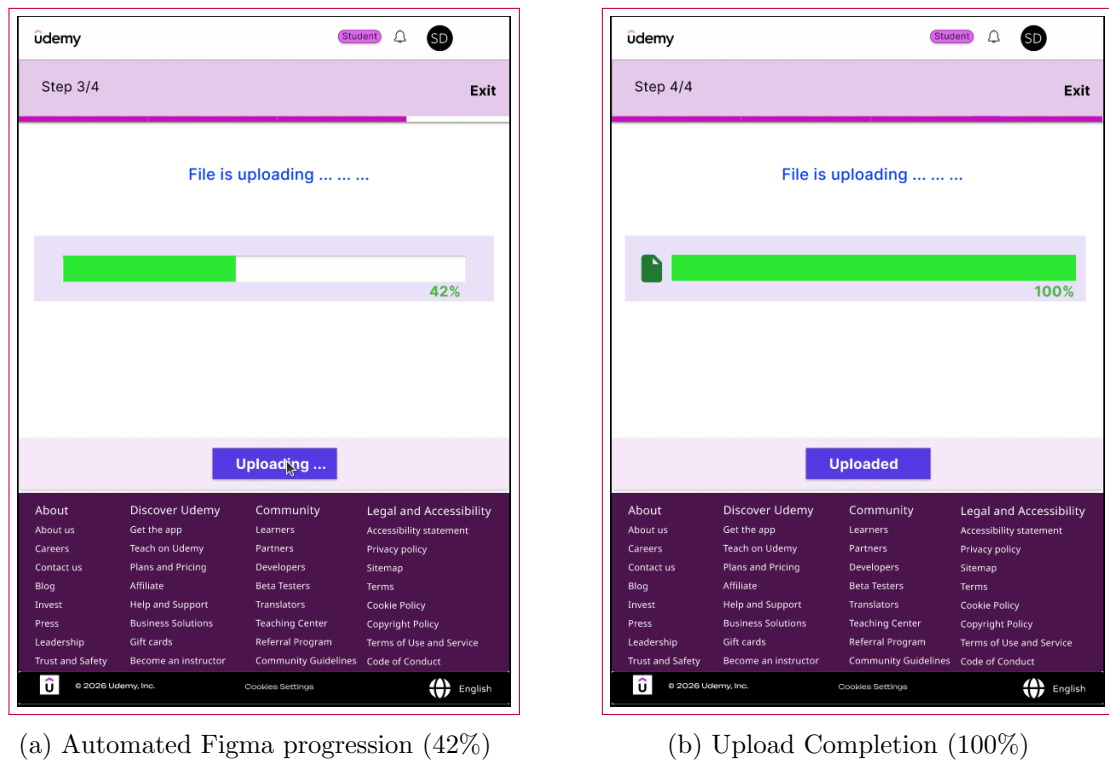
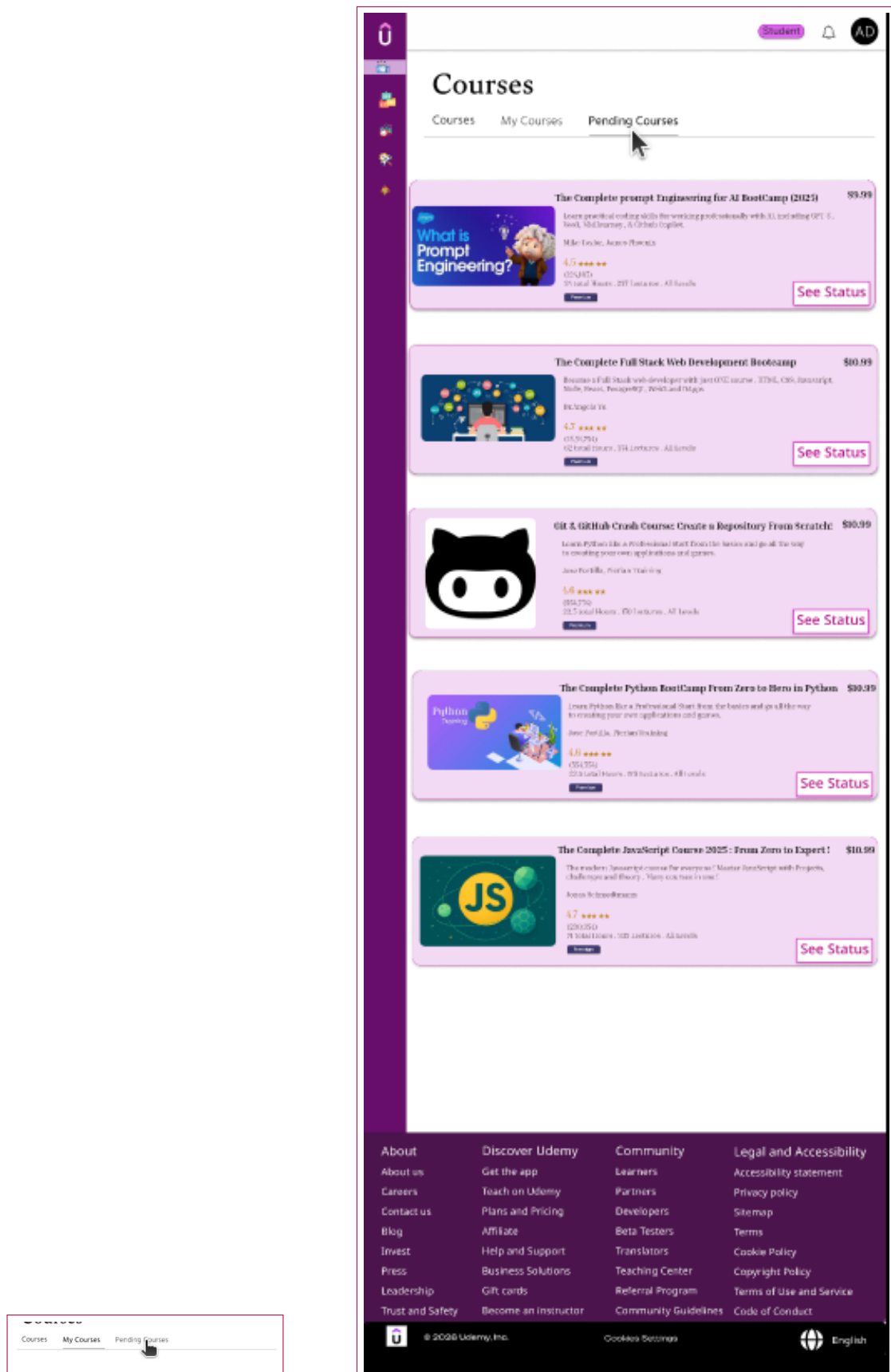


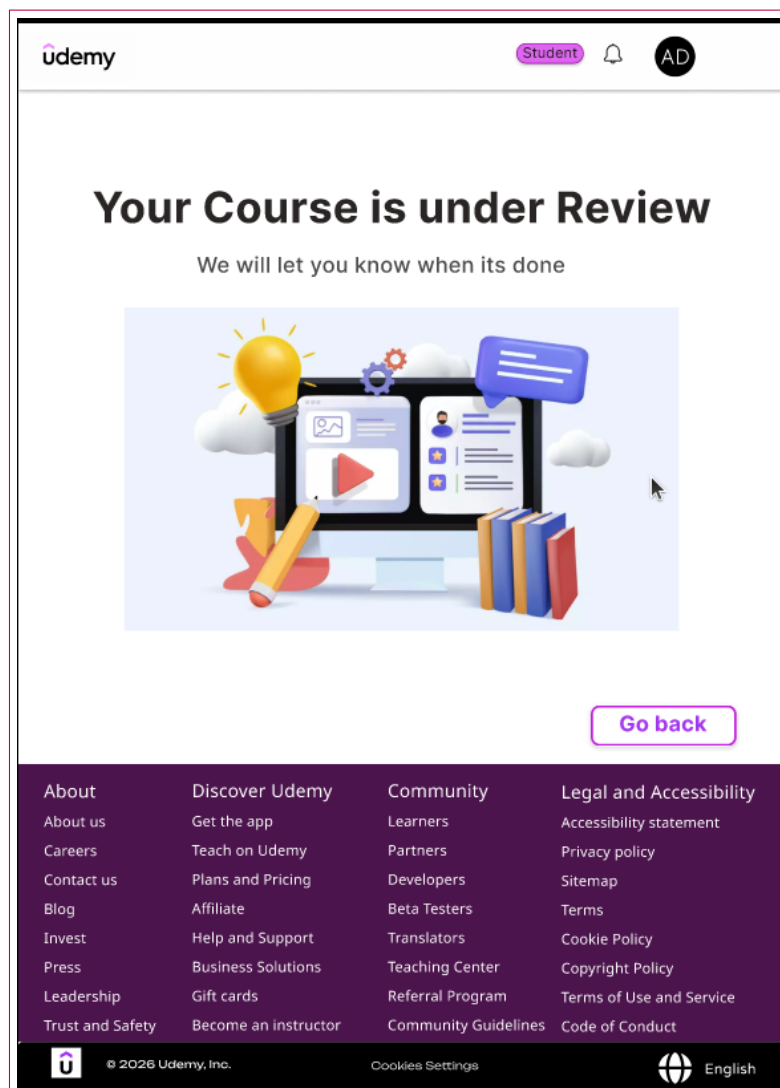
Figure 5.21: Content Upload Process



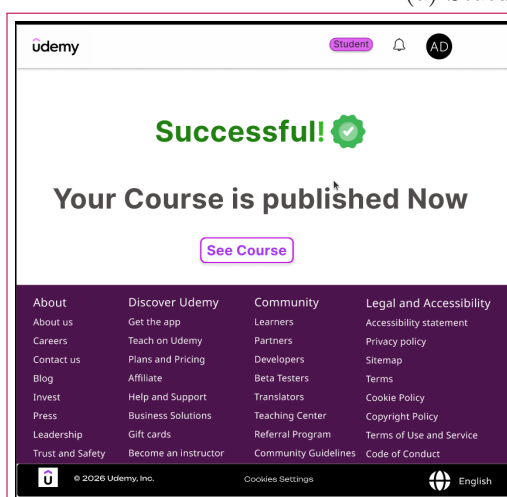
(a) Pending tab

(b) List of courses awaiting approval

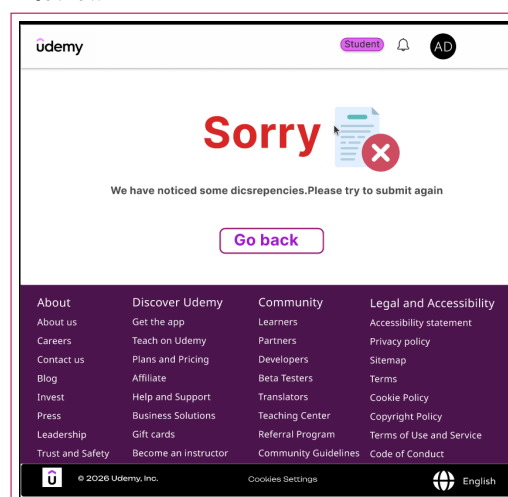
Figure 5.22: Moderation and Pending Courses



(a) Status: Under Review



(b) Status: Published

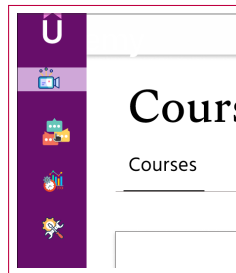


(c) Status: Rejected

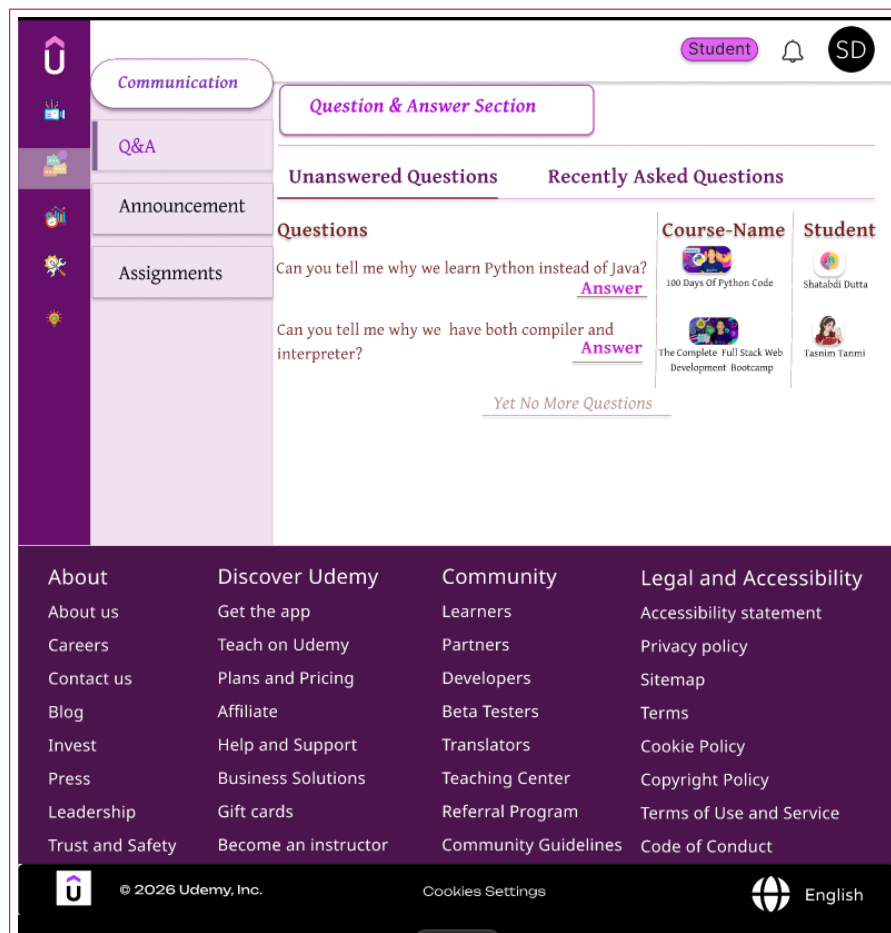
Figure 5.23: Pending Course's Status

5.3.7 Communication Hub, Performance Analytics, and Tools

Fostering a two-way connection is vital for an online learning platform. Instructors have dedicated spaces to interact with students and measure their success using detailed graphical analytics.

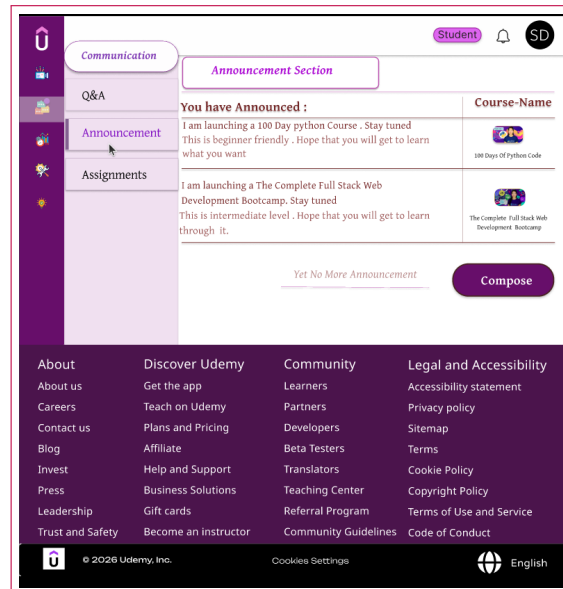


(a) Communication icon

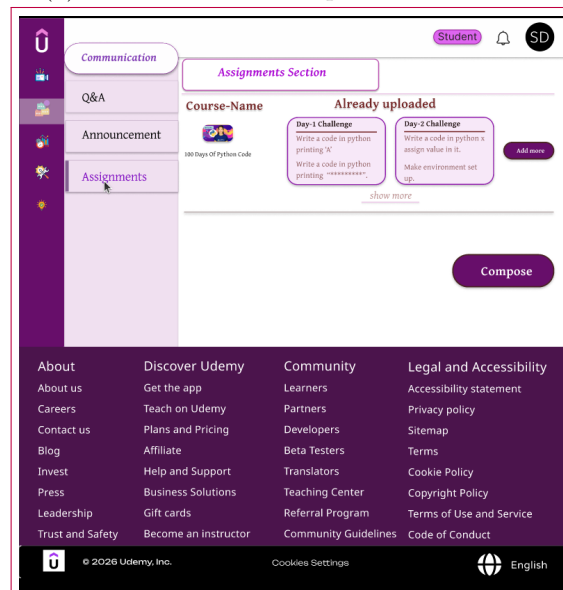


(b) Q&A Dashboard for answering queries

Figure 5.24: Instructor Communication Hub - Question & answer



(a) Announcements composition interface

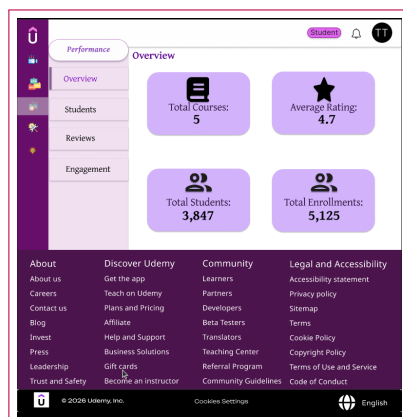


(b) Assignments tracking section

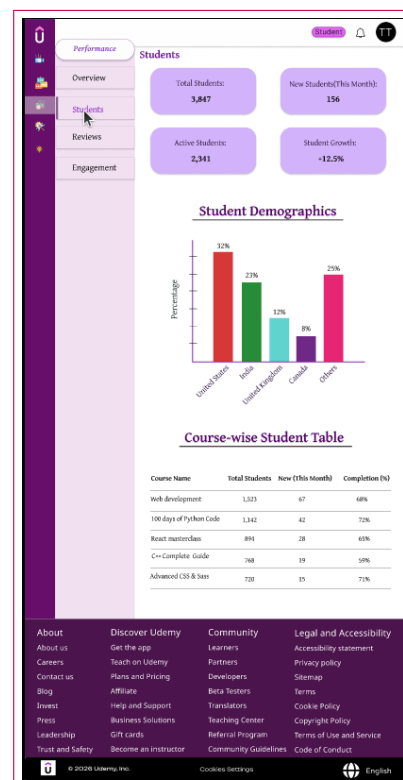
Figure 5.25: Instructor Communication Hub - announcement and assignment



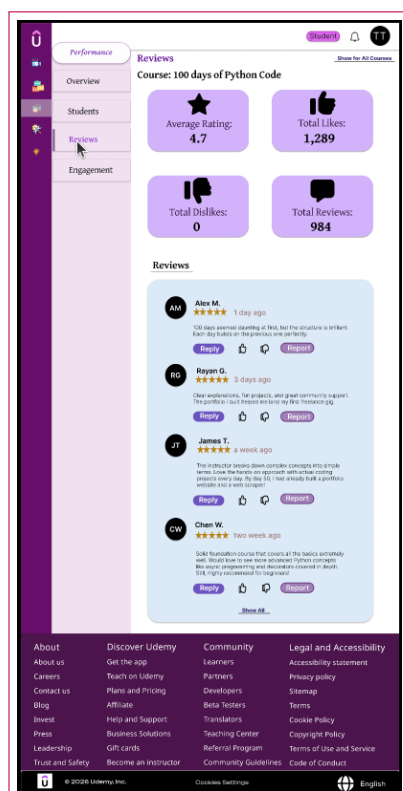
(a) Analytics icon



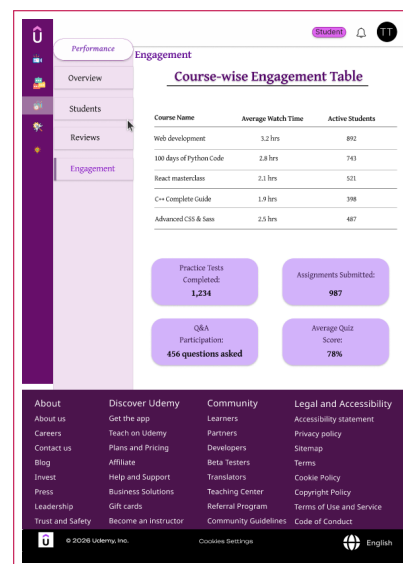
(b) Overview: High-level enrollment and rating stats



(c) Student demographics and course breakdowns

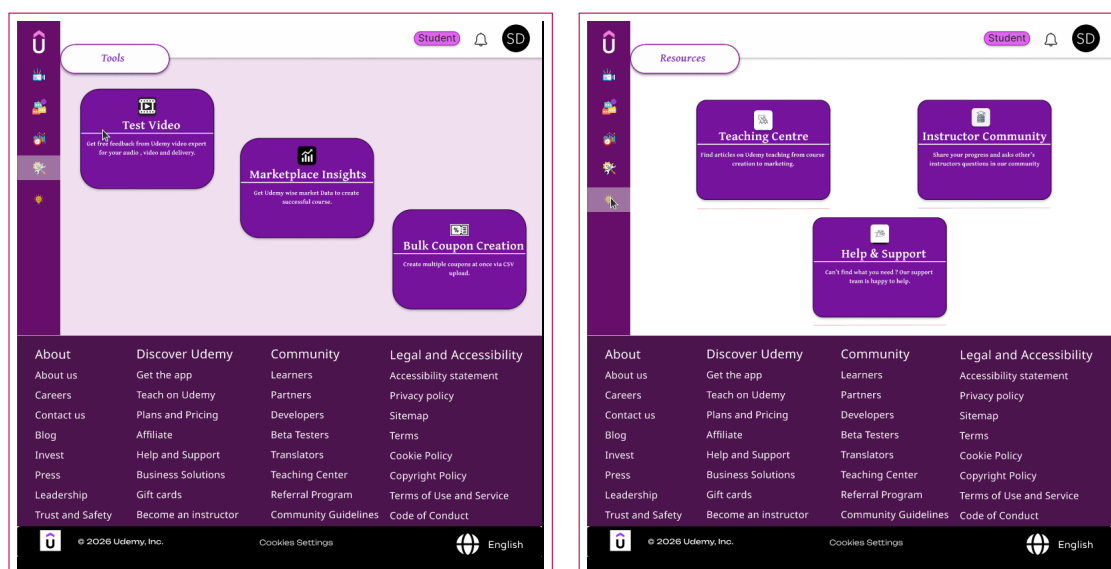


(d) Centralized course reviews tracker



(e) Detailed Engagement metrics and watch times

Figure 5.26: Performance and Analytics Dashboards



(a) Instructor Tools: Video Testing and Market Insights

(b) Instructor Resources: Community and Help Center

Figure 5.27: Platform Tools and Support Resources

5.4 Conclusion

The mock UI presented in this chapter successfully establishes a comprehensive, user-centric design for our Course Management System. By explicitly mapping out 77 detailed navigation flows for both learners and educators, we have visually addressed the core system requirements—ranging from dynamic video delivery and automatic progress tracking to in-depth performance analytics and two-way communication tools. The iterations made based on presentation feedback further refined the visual hierarchy and usability of the platform. Ultimately, these Figma mockups provide a robust, unambiguous, and highly actionable foundation for the subsequent frontend implementation phase of the project.

Chapter 6

Collaboration Diagram

Abid r part

Chapter 7

Sequence Diagram

Presented by: Arpita Dhar (2105137)

7.1 Overview

This chapter presents the sequence diagrams for the UdemY Course Management System. The diagrams illustrate the interaction between different objects and components of the system over time for each core use case.

The following sequence diagrams are included:

- Create Course & Organize Structure
- Upload Video Content
- Upload Learning Materials
- Browse and Enroll in Course
- Watch Video and Progress Tracking
- Resource Access and Download
- Course Review & Rating System
- Instructor Analytics Dashboard
- Admin Review Moderation

7.2 Changes Made

No feedback was provided during the presentation. All diagrams remain unchanged from the original submission.

7.3 Diagrams

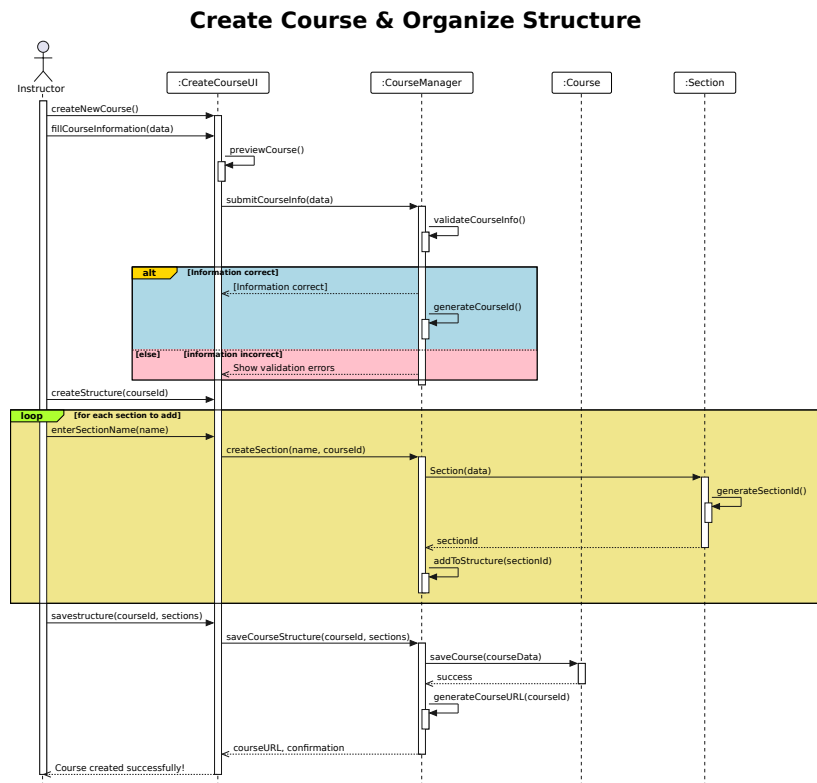


Figure 7.1: Create Course & Organize Structure

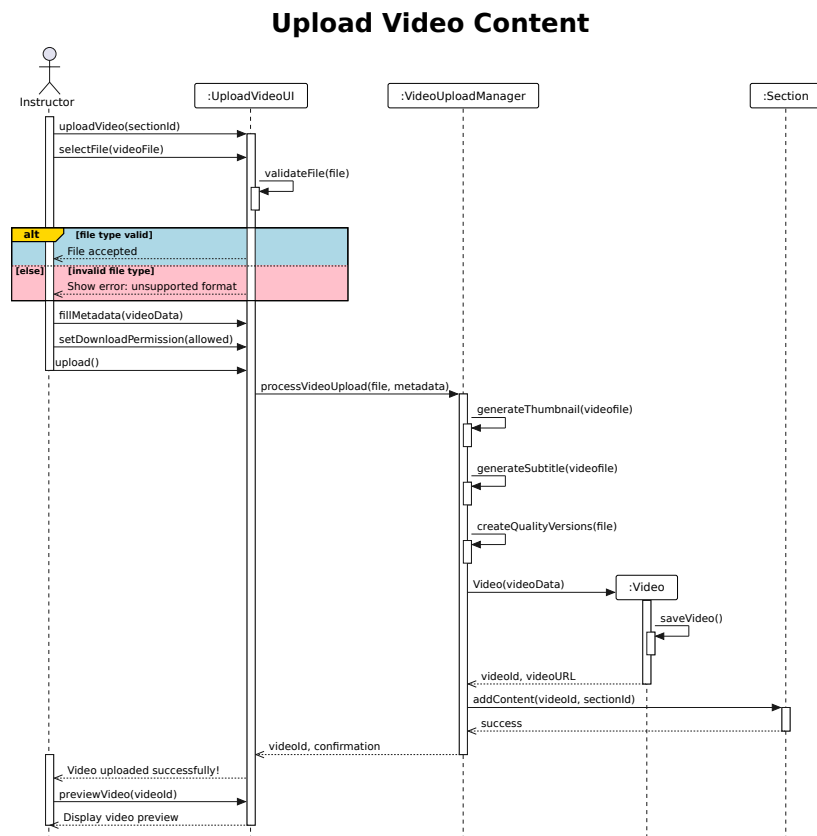


Figure 7.2: Upload Video Content

Upload Learning Materials

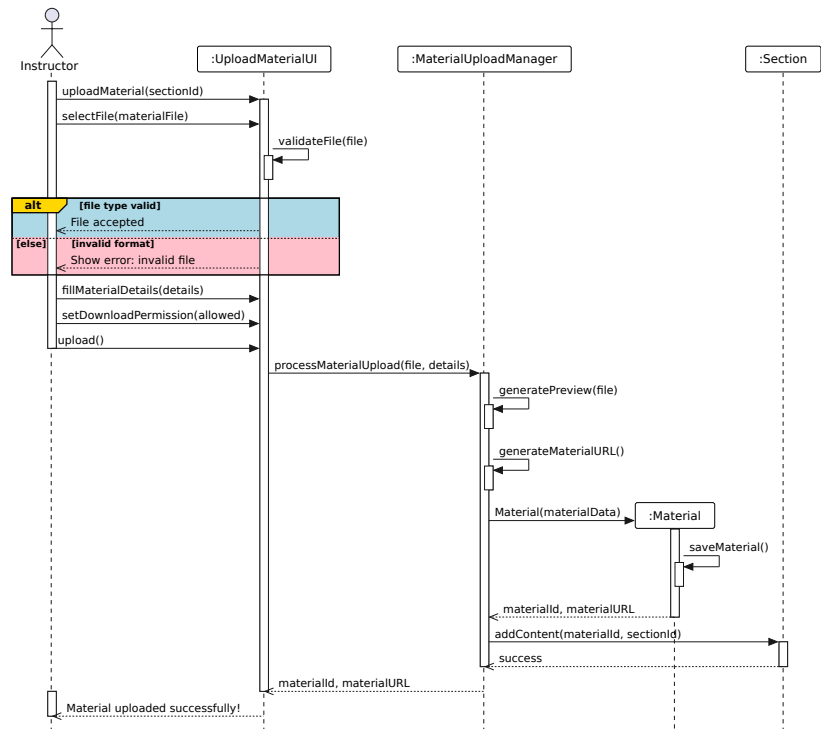


Figure 7.3: Upload Learning Materials

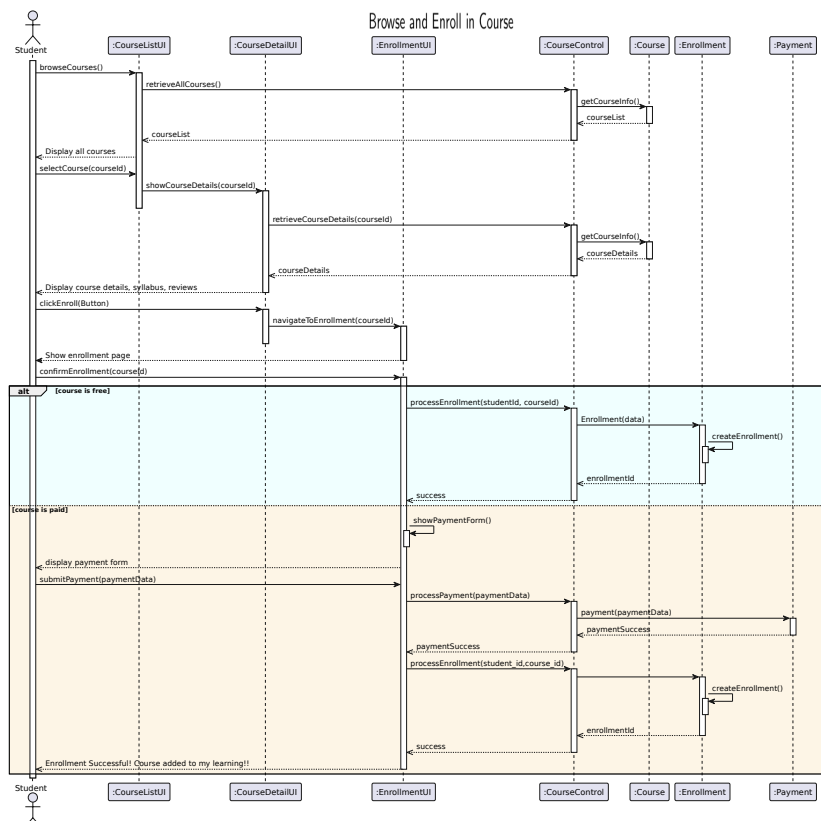


Figure 7.4: Browse and Enroll in Course

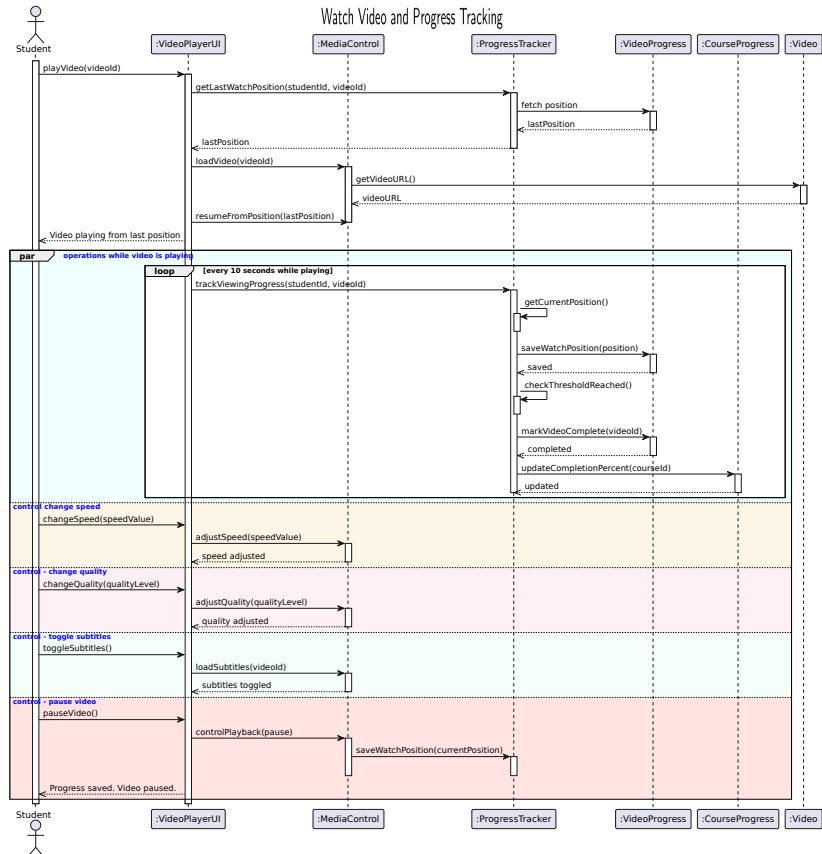


Figure 7.5: Watch Video and Progress Tracking

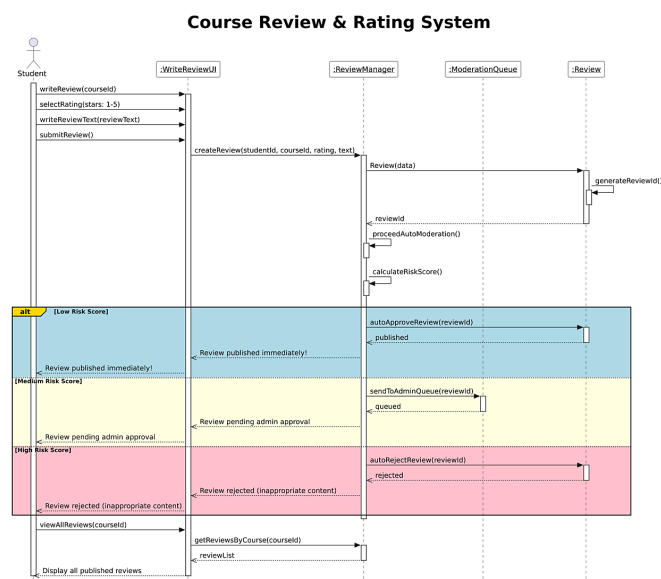


Figure 7.6: Course Review & Rating System

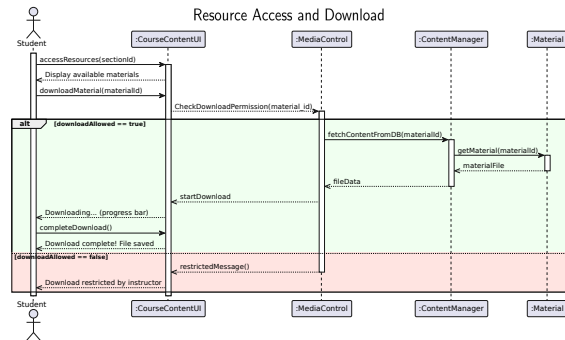


Figure 7.7: Resource Access and Download

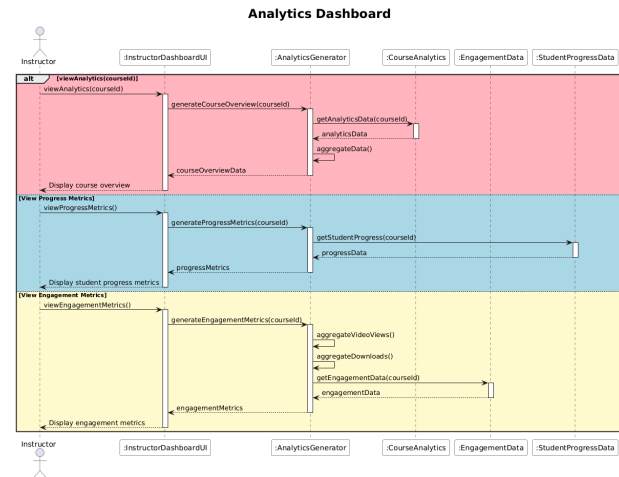


Figure 7.8: Instructor Analytics Dashboard

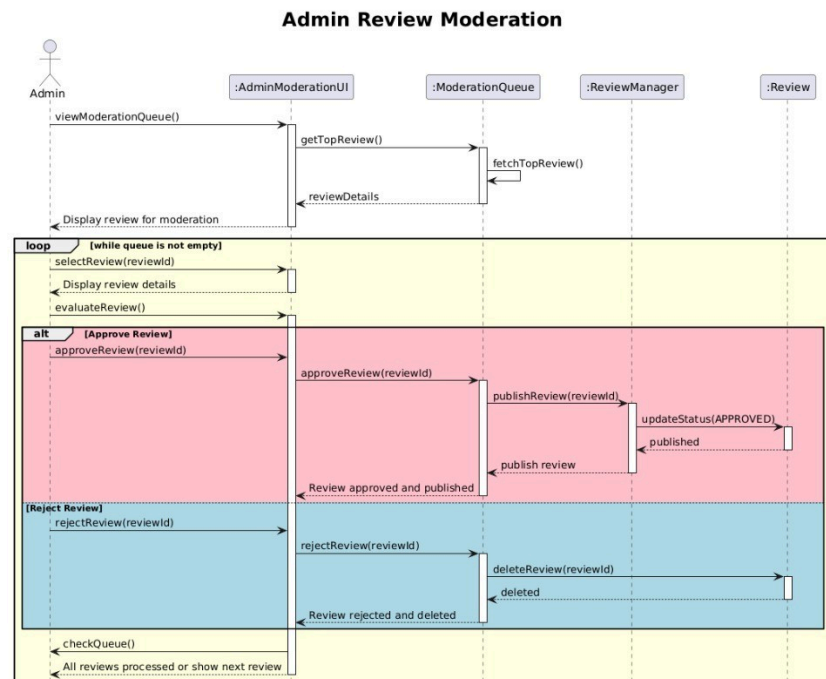


Figure 7.9: Admin Review Moderation

Chapter 8

State Diagram

Presented by: Arpita Dhar (2105137)

8.1 Overview

This chapter presents the state diagrams for the Udemy Course Management System. The diagrams illustrate the various states of key entities in the system and the transitions between those states.

The following state diagrams are included:

- Course Creation
- Student Enrollment Lifecycle
- Video Lecture Lifecycle
- Course Review & Moderation

8.2 Changes Made

During the review of the submitted diagrams, one correction was identified in the **Course Creation State Diagram**. In the original diagram, a `deleteCourse()` transition was incorrectly drawn from the **Saving Course Structure** state looping back to itself. This was a drawing mistake and has been corrected in the updated version — the `deleteCourse()` transition now correctly leads to the terminal state, consistent with the behavior shown in all other states where course deletion results in termination.

8.3 Diagrams

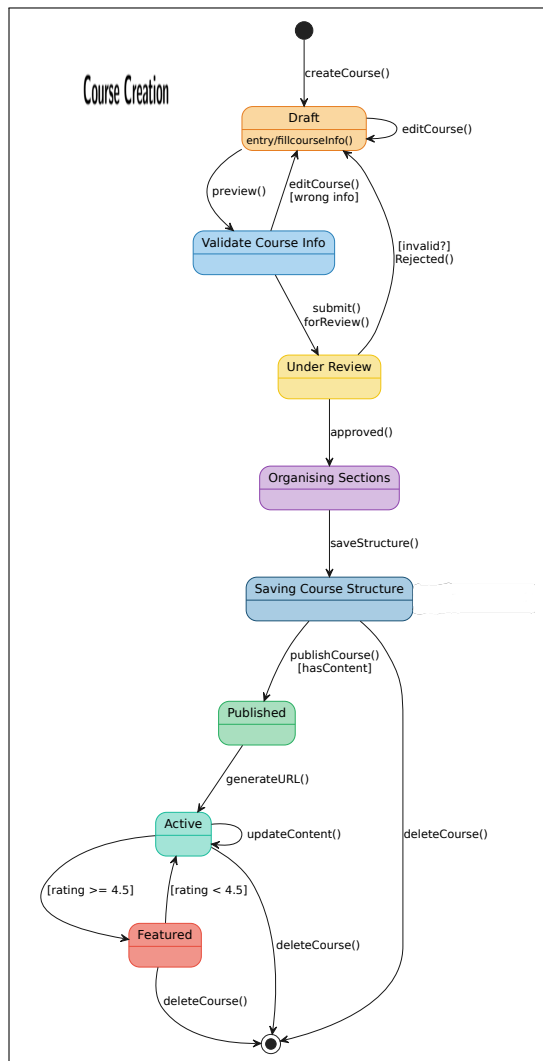


Figure 8.1: Course Creation State Diagram

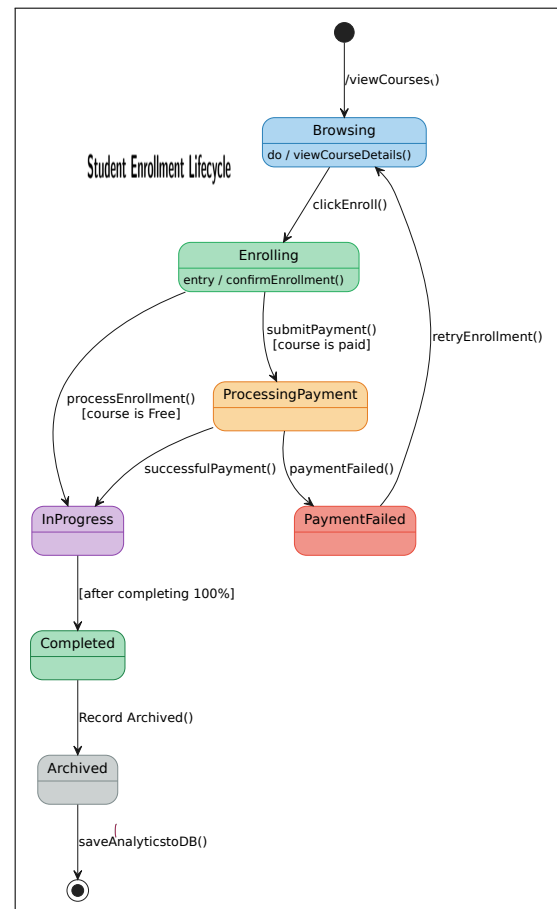


Figure 8.2: Student Enrollment Lifecycle

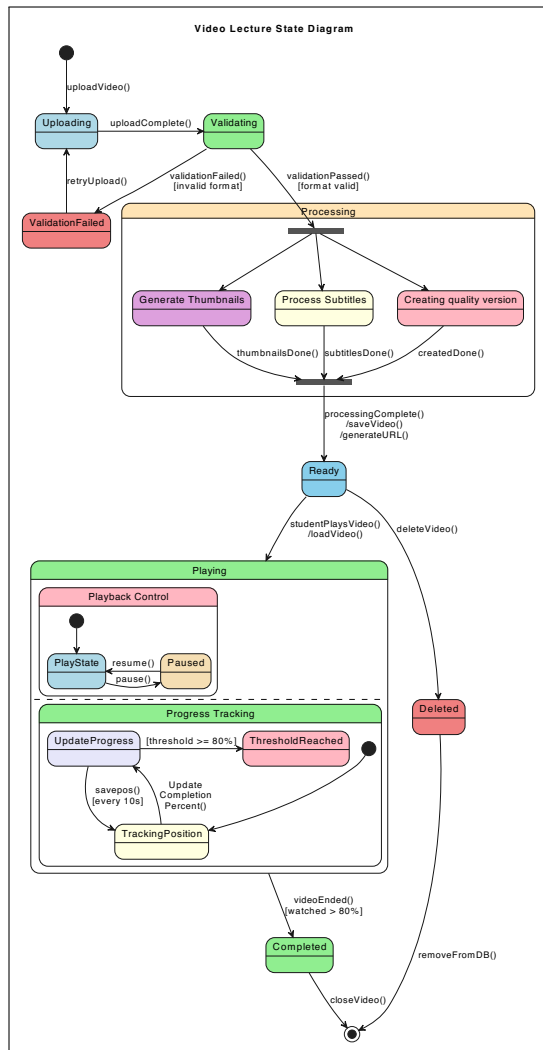


Figure 8.3: Video Lecture Lifecycle

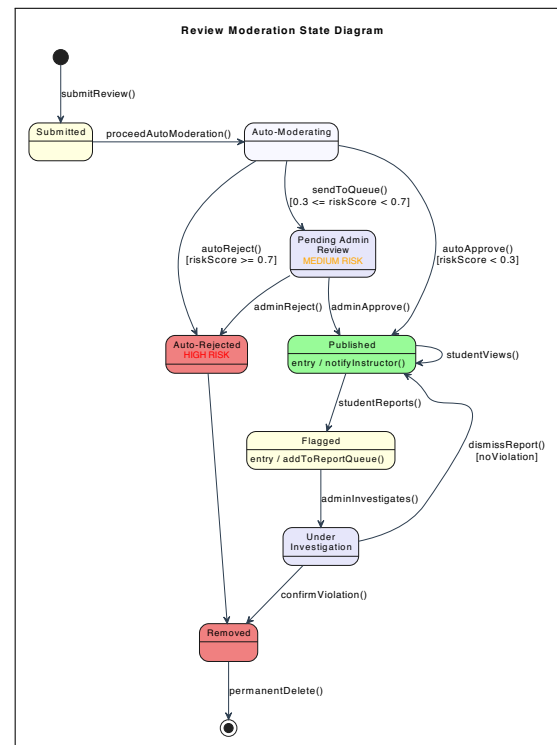


Figure 8.4: Course Review & Moderation

Chapter 9

API Documentation

Presented By: MD. Yousuf Niaz (2105125)

9.1 Overview

Udemy is a leading global learning platform with over 210,000 courses. This documentation details a focused API subset organized into the following modules:

Module	Description
Authentication	OTP-based signup, login, and token management.
Course	Listings, status tracking, and creation.
Student	Dashboards, enrollments, and learning history.
Instructor	Course management, analytics, and earnings.
Popular	Trending topics and top-rated instructors.
Others	Plans, categories, and notification systems.

9.2 Changes Made

- **v1.2:** Added OAuth2 support.
- **v1.1:** Fixed rate-limiting headers.
- **v1.0:** Initial release of the documentation.

9.3 Modules

9.3.1 Authentication

9.3.1.1 Register User & Send OTP

POST /auth/signup – Creates a pending account and triggers a 6-digit OTP.

Headers & Parameters	Responses
Headers: Content-Type: <code>application/json</code> X-Request-ID: <i>string (optional)</i> Body (JSON): email: <i>string (req)</i> full_name: <i>string (req)</i> role: <i>student — instructor</i> marketing_opt_in: <i>bool</i>	202 Accepted: OTP sent. 400 Bad Request: Malformed fields. 409 Conflict: Email already exists. 422 Unprocessable: Validation failed. 429 Rate Limit: Too many requests. 500/503: Server error/maintenance.

Role Logic: *Student* gets learning dashboard; *Instructor* gets creation/analytics tools.

9.3.1.2 Verify OTP & Activate

POST /auth/signup/verify-otp – Validates OTP and issues JWT session tokens.

Request Details	Responses
Headers: Content-Type: <code>application/json</code> Body (JSON): email: <i>string (req)</i> otp: <i>6-digit string (req)</i>	201 Created: Account active. Tokens issued. 400 Bad Request: Missing email/otp. 401 Unauthorized: Incorrect OTP. 410 Gone: OTP expired. Restart signup. 429 Rate Limit: Limit exceeded. 500 Error: Unexpected failure.

9.3.1.3 Initiate Login & Send OTP

POST /auth/login – Triggers a 6-digit OTP to the registered email.

Request Details	Responses
Headers: Content-Type: application/json Body (JSON): email: <i>string (req)</i>	202 Accepted: OTP sent if email exists. 400 Bad Request: Missing/malformed email. 429 Rate Limit: Rate limit hit for email. 500/503: Server failure or maintenance.

Security Note: Response is identical regardless of email existence to prevent enumeration attacks.

9.3.1.4 Verify OTP & Complete Login

POST /auth/login/verify-otp – Validates OTP and issues JWT session tokens.

Request Details	Responses
Headers: Content-Type: application/json Body (JSON): email: <i>string (req)</i> otp: <i>6-digit string (req)</i>	200 OK: Login successful. Tokens issued. 401 Unauthorized: OTP incorrect. 410 Gone: OTP expired. Request new. 429 Rate Limit: Limit exceeded. 500 Error: Unexpected server failure.

Redirection Logic: *student* → learning dashboard; *instructor* → creator dashboard.

9.3.1.5 Refresh Access Token

POST /auth/token/refresh – Rotates the refresh token and issues a new access token.

Request Details	Responses
Body (JSON): refresh_token: <i>string (req)</i>	200 OK: New access/refresh pair issued. 401 Unauthorized: Token invalid or used. 410 Gone: Expired. Re-login required. 500 Error: Unexpected server failure.

9.3.1.6 Logout & Invalidate Tokens

POST /auth/logout – Invalidates tokens server-side. Client must clear local storage.

Request Details	Responses
Headers: Authorization: Bearer <token> Body (JSON - Optional): refresh_token: <i>string</i> <i>(Include for full session cleanup)</i>	204 No Content: Logged out. Tokens invalidated. 401 Unauthorized: Missing or invalid access token. 500 Error: Unexpected server failure.

9.3.2 Course

9.3.2.1 List Courses

GET /courses – Returns a paginated, filtered list of courses.

Query Parameters	Responses
Filters: <i>category: slug (e.g., development, business)</i> <i>filter: most-popular — trending — new</i> Pagination: <i>limit: int (1-50, default 3)</i> <i>offset: int (skip count)</i> <i>Ex: ?limit=3&offset=6 (Page 3)</i>	200 OK: Course list returned. 400 Bad Request: Invalid parameter. 429 Rate Limit: Limit exceeded. 500/503: Server failure/maintenance.

9.3.2.2 Create a New Course

POST /courses – Creates a pending course (Instructor role required).

Request Details	Responses
Headers: Auth: Bearer <instructor_token> Content-Type: application/json Body (JSON): <i>title: string (5-150 chars)</i> <i>category: slug (dev, business, etc.)</i> <i>weekly_time_commitment:</i> <i>(0-2h — 2-4h — 5h+ — undecided)</i>	201 Created: Course set to pending. 400/422: Malformed or validation failed. 401 Unauthorized: Invalid token. 403 Forbidden: Not an instructor. 429 Rate Limit: Limit exceeded. 500 Error: Server failure.

Note: Instructor ID is extracted from the token. Drafts are invisible to students until published.

9.3.2.3 Get Course Details

GET /courses/{course_id} – Fetches full course metadata, instructor profile, and student reviews.

Path Parameters	Responses
Required: <i>course_id: string</i> <i>(Unique identifier for the course)</i>	200 OK: Detailed course info returned. 429 Rate Limit: Limit exceeded. 500 Error: Unexpected server failure. 503 Service: Maintenance/Overloaded.

9.3.2.4 Get Course Approval Status

GET /courses/{course_id}/status – Returns the current review state of a course.

Parameters & Logic	Responses
Path Param: <i>course_id: string (prefix crs_)</i> Lifecycle Flow: pending → accepted — rejected <i>(Rejected status includes rejection_reason)</i>	200 OK: Status returned. 401 Unauthorized: Invalid token. 403 Forbidden: Not owner/admin. 404 Not Found: ID not found. 429 Rate Limit: Limit exceeded. 500 Error: Server failure.

Definitions: *Pending:* Awaiting review; *Accepted:* Visible to students; *Rejected:* See reason for details.

9.3.2.5 Upload Course Media

POST /courses/{course_id}/upload – Uploads lecture videos or downloadable materials.

Params & Content	Responses
Path Param: <i>course_id</i> Query Param: <i>file_type: video — material</i> Headers: Content-Type: multipart/form-data Limits: <i>video:</i> .mp4, .mov, .avi, .mkv (Max 4GB) <i>material:</i> .pdf, .zip, .pptx, .docx (Max 100MB)	200/201: Success (Replenish/New). 401/403: Auth/Ownership error. 413: Content too large. 415: Unsupported media type. 422: Corrupted or unreadable file. 500: Server failure.

Note: Replaces existing file if the slot (*file_type*) is already occupied.

9.3.3 Student

9.3.3.1 Get Dashboard Summary

GET /student/dashboard – Returns high-level metrics for the authenticated student.

Request Details	Responses
Headers: Auth: Bearer <student_token> Logic: <i>Student ID is extracted from the JWT. Only accessible to users with <code>role: student</code>.</i>	200 OK: Returns display name, enrolled counts, and last accessed course. 401 Unauthorized: Missing or invalid token. 403 Forbidden: User does not have student role. 429 Rate Limit: Limit exceeded. 500 Error: Unexpected server failure.

UI Note: Counts (*notification_count*, *cart_count*) are intended for rendering badge numbers on UI icons.

9.3.3.2 Get Enrolled Courses

GET /student/courses – Returns all courses the student is currently enrolled in.

Request Details	Responses
Headers: Auth: Bearer <student_token> Sorting: Results are automatically sorted by <i>last accessed date</i> (most recent first).	200 OK: List of enrolled courses returned. 401 Unauthorized: Missing or invalid token. 403 Forbidden: Not a student. 500 Error: Unexpected server failure.

9.3.3.3 Add or Remove Course from Learning List

PATCH /student/my-learning – Partially updates the student's enrolment list. Use **action: add** to enroll, **action: remove** to unenroll.

Parameters	Responses
Headers: Authorization: <i>string (Bearer token) required</i> Content-Type: <i>application/json required</i> Body (JSON): action: <i>string enum required</i> "add" – Enroll in a course "remove" – Remove a course from the learning list course_id: <i>string required</i> <i>ID of the course to add or remove</i>	200 OK: Learning list updated successfully. 400 Bad Request: Missing or invalid action/course_id. 401 Unauthorized: Missing or invalid token. 403 Forbidden: Not the resource owner or wrong role. 404 Not Found: course_id does not exist. 409 Conflict: Course already in list (add) or not in list (remove). 500 Internal Server Error: Unexpected server failure.

9.3.3.4 Get Cart Contents

GET /student/cart – Returns all courses currently in the student’s cart with price details.

Parameters	Responses
Headers: Authorization: <i>string (Bearer token) required</i>	200 OK: Cart contents returned. 401 Unauthorized: Missing or invalid token. 403 Forbidden: Not the resource owner or wrong role. 500 Internal Server Error: Unexpected server failure.

9.3.3.5 Add a Course to Cart

POST /student/cart – Adds the specified course to the student’s cart.

Parameters	Responses
Headers: Authorization: <i>string (Bearer token) required</i> Content-Type: <i>application/json required</i> Body (JSON): course_id: <i>string required</i> <i>ID of the course to add</i>	200 OK: Course added to cart (or already present). 401 Unauthorized: Missing or invalid token. 403 Forbidden: Not the resource owner or wrong role. 404 Not Found: Course does not exist. 409 Conflict: Course is already enrolled. 500 Internal Server Error: Unexpected server failure.

9.3.3.6 Remove a Course from Cart

DELETE /student/cart/{course_id} – Removes the specified course from the student’s cart.

Parameters	Responses
Path Param: <i>course_id: string required</i> <i>ID of the course to remove</i> Headers: <i>Authorization: string (Bearer token) required</i>	200 OK: Course removed from cart. 401 Unauthorized: Missing or invalid token. 403 Forbidden: Not the resource owner or wrong role. 404 Not Found: Course is not in this student's cart. 500 Internal Server Error: Unexpected server failure.

9.3.3.7 Load Checkout Page Data

GET /student/checkout – Loads all data needed to render the checkout page (cart items, totals, discounts, saved payment details).

This endpoint only prepares the order. It does not place it.

Parameters	Responses
Headers: <i>Authorization: string (Bearer token) required</i>	200 OK: Checkout data returned. 401 Unauthorized: Missing or invalid token. 403 Forbidden: Not the resource owner or wrong role. 422 Unprocessable Content: Cart is empty, cannot checkout. 500 Internal Server Error: Unexpected server failure.

9.3.4 Instructor

9.3.4.1 Get Instructor Profile

GET /instructor/profile – Returns the instructor's full profile — bio, stats, social links, and a list of their courses.

Request Details	Responses
Headers:	200 OK: Instructor profile returned.
Authorization: Authorization: <i>string</i>	401 Unauthorized: Missing or expired access token.
<i>(Bearer token)</i>	403 Forbidden: Wrong resource owner or wrong role.
	404 Not Found: No instructor found with this ID.
	500 Internal Server Error: Unexpected server failure.

9.3.4.2 Get Q&A Questions

GET /instructor/qa – Returns Q&A questions across all of the instructor’s courses. Use the **filter** query param to narrow results.

Request Details	Responses
Headers:	200 OK: Q&A list returned.
Authorization: <i>string</i> (<i>Bearer</i> <i>token</i>)	400 Bad Request: Invalid filter value.
Query Parameters:	401 Unauthorized: Missing or expired access token.
filter (<i>enum: unanswered, recent</i>)	403 Forbidden: Not the resource owner or wrong role.
Default: recent	500 Internal Server Error: Unexpected server failure.

9.3.4.3 Get All Course Announcements

GET /instructor/announcements – Returns all announcements published by the instructor across all their courses, sorted by published date (newest first).

Request Details	Responses
Headers:	200 OK: Announcements returned.
Authorization: Bearer	401 Unauthorized: Missing or expired access token.
<instructor_token> Logic:	403 Forbidden: Wrong resource owner or wrong role.
<i>Instructor ID is extracted from the JWT.</i>	500 Internal Server Error: Unexpected server failure.
<i>Only accessible to users with role: instructor.</i>	

9.3.4.4 Get All Course Assignments

GET /instructor/assignments – Returns all assignments across the instructor’s courses. Each assignment includes day-by-day content (day-1, day-2, ...).

Request Details	Responses
Headers: Authorization: Bearer <instructor_token> Logic: <i>Instructor ID is extracted from the JWT. Only accessible to users with role: instructor.</i>	200 OK: Assignments returned. 401 Unauthorized: Missing or expired access token. 403 Forbidden: Wrong resource owner or wrong role. 500 Internal Server Error: Unexpected server failure.

Chapter 10

Feature Implementation

Implemented Feature Name : Automatic Progress Tracking

10.1 Description

The **Automatic Progress Tracking** feature introduces an intelligent, system-driven mechanism that continuously monitors and records each student's learning activity without requiring any manual input. This eliminates administrative burden while ensuring accurate, real-time visibility into course engagement and completion status.

The system employs **auto-tracking** functionality whereby a lecture is automatically marked as complete once a student meets a predefined viewing threshold, ensuring that progress data reflects actual engagement rather than mere access. Alongside this, an **Analytics Dashboard** provides both students and instructors with detailed insights, including completion percentages and milestone achievements — enabling instructors to identify at-risk students early and allowing learners to stay informed of their own progress throughout the course.

Together, these components form a cohesive progress management system that enhances accountability and supports data-informed decision-making for all stakeholders.

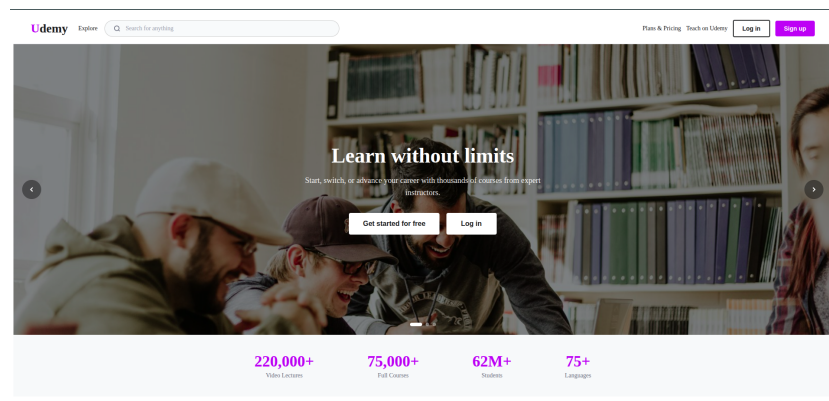


Figure 10.1: Homepage

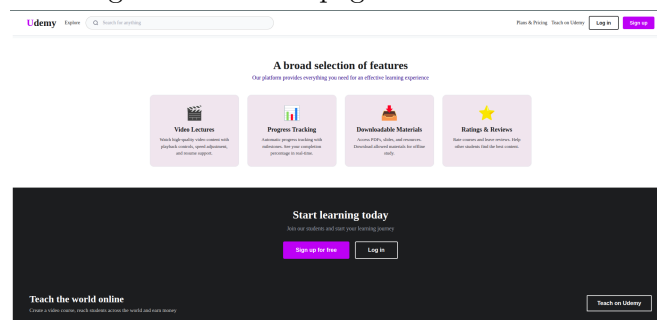


Figure 10.2: Boarder Selection of Features

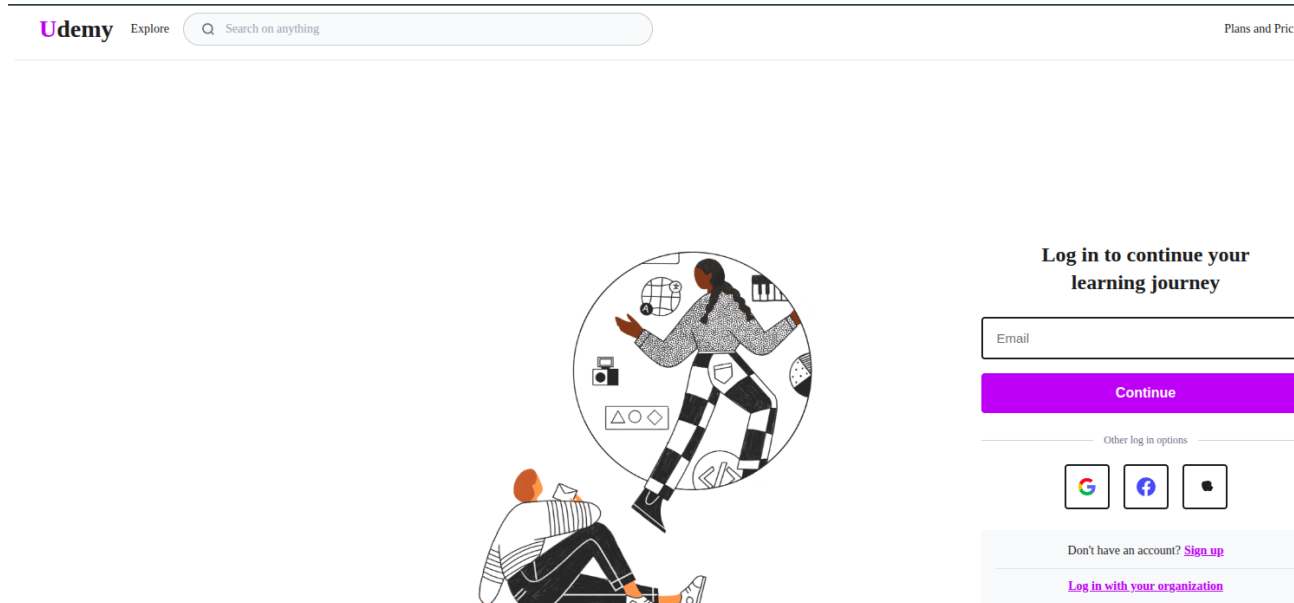


Figure 10.3: Login Page

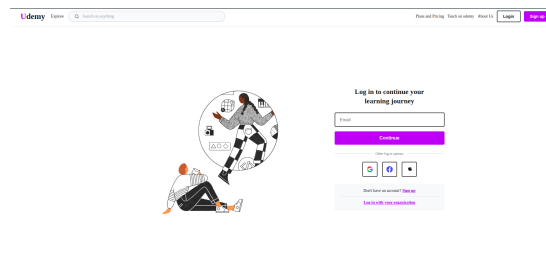


Figure 10.4: Login Page

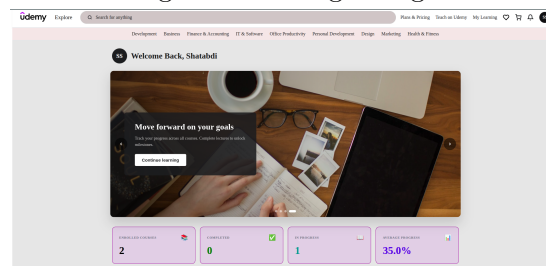


Figure 10.5

Chapter 11

Conclusion
